Human 1.0: Water:Fat in the breast

Summary

Human 1.0 shows that the difference between NORMAL and CANCER breasts is in the **water-rich scans**. Measured there, a basic analysis points to the CANCER breast rather than the NORMAL one, matching synchrotron and ex-vivo data on the water-rich side.

What we expect: a normal breast is mostly fat with some glandular/fibrous tissue. Cancer replaces fat with cells and remodelled fibrous tissue, which hold more water. Each breast gets one number, its **Water:Fat ratio**: the area of its water peak divided by the area of its fat peak, from two fitted Gaussians. Comparing a patient's two breasts, the **CANCER breast should have the higher Water:Fat**.

Data: 258 patients analysed with Human 1.0 (NORMAL, BENIGN or CANCER); **251** used, after leaving out 7 with a large left/right intensity imbalance (Appendix I).

1. The difference is in the water-rich scans

In the **fattiest 15%** of samples the three cohorts look the same; in the **wettest 15%** they separate NORMAL < BENIGN < CANCER. The information is in the water-rich tissue — denser breasts are better for us; mammography's weakness is our strength (Chapter 1).

2. It matches other data

That wettest-15% ordering matches synchrotron and Keele ex-vivo data. A fixed water-peak threshold for single-point scans (Appendix C) also keeps the wet tissue; the paired test there supports CANCER vs NORMAL (Chapter 2).

3. Rigorous tumour targeting is not needed

The tumours are small and only a few beams are placed approximately, yet the signal holds where the beam goes through glandular tissue: changes appear wide-reaching in the glandular breast, not confined to the tumour (Chapter 3).

4. Mapping for Human 2.0 Proposal

Scan a grid over the whole breast and select the wettest tissue after the scan. The water-peak rule (Appendix C) could be a stopgap on Human 1.0; mapping is the recommendation (Chapter 4). A simulated half breast at 5 mm steps, one second per point, finishes in 2 min 46 s (Chapter 5).

A pilot result: 18 CANCER/NORMAL patients, 95% CI 59%–96%, and mostly dense breasts. All limitations are in Chapter 6.

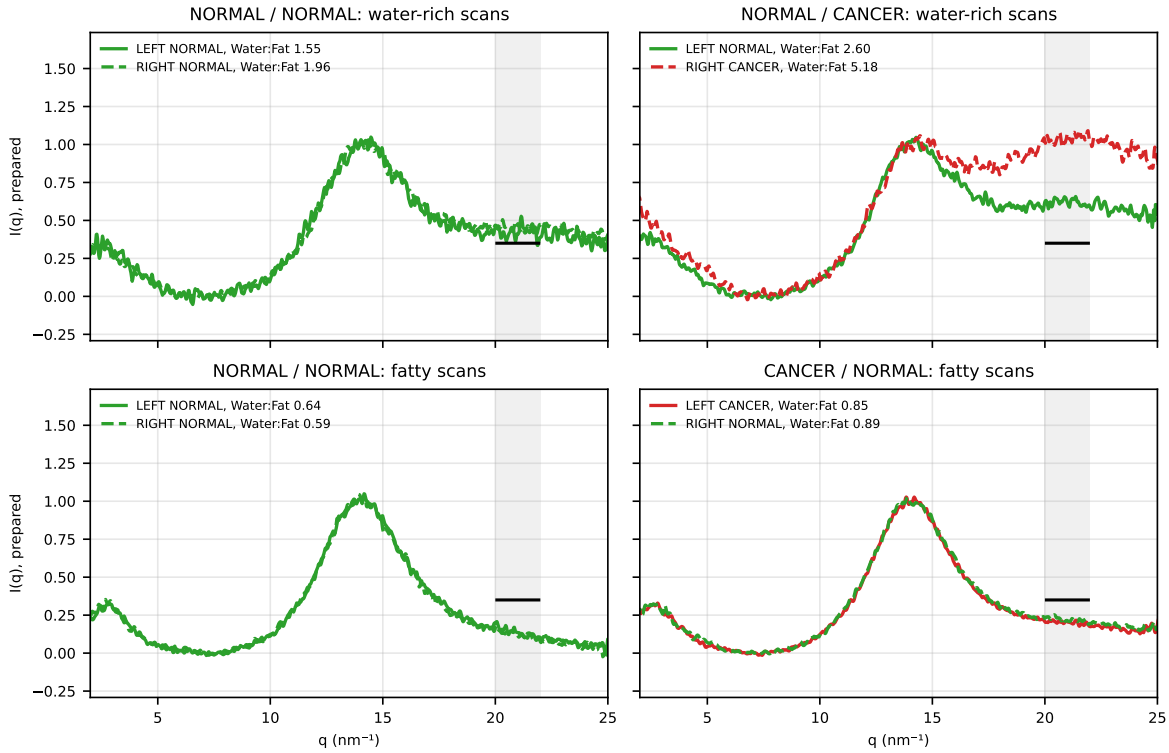


Figure 1: **Top**: water-rich scans (usable). **Bottom**: fatty scans the rule rejects. Each panel is one patient; each curve the mean of that breast's scans of that kind. On water-rich scans the CANCER breast stands out; on fatty scans NORMAL/NORMAL and NORMAL/CANCER look alike.

1 The difference between cohorts is in the water-rich scans

> **Finding 1**
>
> In the **fattiest 15%** of samples the three cohorts look the same. In the **wettest 15%** they separate: NORMAL 0.81 < BENIGN 0.96 < CANCER 1.07 (water peak height). The information is in the water-rich tissue.

Every prepared sample from Nova 250 onwards (intensity-imbalanced patients left out) is shown in Figure 1.1: by Nova number, and ordered by **water peak height** (the mean of the prepared curve over q = 20-22; preparation sets the fat peak to 1, Appendix A). The wettest and fattiest 15% are marked.

Figure 1.2 averages those two bands by cohort. The fattiest 15% overlaps for all three labels (water peak about 0.12). The wettest 15% pulls apart. Most breast cancers arise in the glandular tissue [1] [3], and tumours remodel the fibrous tissue around them [5], so a scan through fat sees little of it.

Denser is better for us. Dense breasts hold more glandular, water-rich tissue — exactly where this signal lives. That is where mammography is weakest (density can hide masses) [8] [9], so mammography's weakness is our strength: the same tissue that hurts X-ray contrast is what XRD needs. For **mapping**, keep the wettest points after a whole-breast scan (e.g. the top 15% within each breast) and compare those (Chapter 4).

```
print(f"{len(all_names)} samples from {len(set(map(ne.row_nova, all_names)))} patients, Nova
plt.rcParams["font.size"] = 7
fig, axes = plt.subplots(1, 2, figsize=(8, 3.9))
for ax in axes:
    ax.set_box_aspect(1)
for ax, image_rows in ((axes[0], all_rows), (axes[1], all_rows[all_by_water])):
    covered = np.flatnonzero(np.isfinite(image_rows).all(axis=0))
    columns = slice(covered[0], covered[-1] + 1)
    low, high = np.nanpercentile(image_rows[:, columns], (1, 99))
    left, right = q[columns][[0, -1]]
    ax.imshow(image_rows[:, columns], aspect="auto", cmap=cmap, vmin=low, vmax=high, interpol
              extent=(left, right, len(image_rows), 0))
```

```
    ax.set(xlabel="q (nm ⁻¹)", xlim=(2, right))
    ax.set_xticks([5, 10, 15, 20])
    ax.axvline(14, color="white", ls="--", lw=0.8)
for (start, stop), colour, text in (((0, all_part), TOP_COLOUR, f"top {PART}%"),
                                    ((len(all_rows) - all_part, len(all_rows)), BOTTOM_COLOUR
    axes[1].add_patch(plt.Rectangle((2, start), right - 2, stop - start, fill=False, edgecolo
    axes[1].text(right - 0.4, (start + stop) / 2, text, color="white", ha="right", va="center
axes[0].set_yticks([0.5, len(all_names) - 0.5], [f"Nova {ne.row_nova(all_names[0])}", f"Nova
axes[0].set_ylabel(f"{len(all_names)} samples")
axes[1].set_yticks([])
axes[1].annotate("", xy=(-0.11, 0.9), xytext=(-0.11, 0.1), xycoords="axes fraction",
                 arrowprops=dict(arrowstyle="-|>", color="black", lw=1.2), annotation_clip=F
axes[1].text(-0.11, 0.07, "Fatty", transform=axes[1].transAxes, ha="center", va="top")
axes[1].text(-0.11, 0.93, "Fibrous/\nGlandular", transform=axes[1].transAxes, ha="center", va
fig.subplots_adjust(left=0.09, right=0.98, bottom=0.1, top=0.97, wspace=0.4)
plt.show()
plt.rcParams["font.size"] = 10
```

```
445 samples from 251 patients, Nova 250 onwards
```

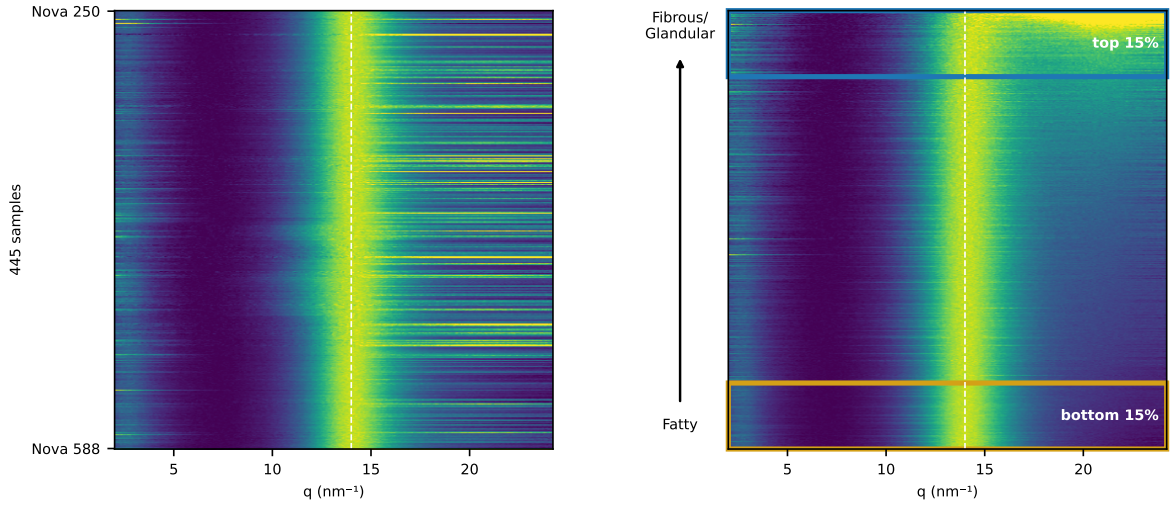


Figure 1.1: Every NORMAL, BENIGN and CANCER sample from Nova 250 onwards (intensity-imbalanced patients left out), after preparation. **Left**: ordered by Nova number, smallest at the bottom. **Right**: ordered by water peak height (mean over q 20-22), highest at the top, with the wettest top 15% (blue) and fattiest bottom 15% (yellow) outlined.

```
covered = np.flatnonzero(np.isfinite(np.vstack([m for parts in part_means.values() for m, _ i
columns = slice(covered[0], covered[-1] + 1)
plt.rcParams["font.size"] = 7
fig, axes = plt.subplots(1, 2, figsize=(8, 3.6), sharey=True)
for ax, part in zip(axes, ("bottom", "top")):
    ax.set_box_aspect(1)
    for label in COHORTS:
        mean, n = part_means[label][part]
        ax.plot(q[columns], mean[columns], color=ne.STATUS_COLOURS[label], lw=1.6,
                label=f"{label}: {n} samples, water peak {part_water[label][part]:.2f}")
    ax.axvline(14, color="black", ls="--", lw=0.8)
    ax.set(title=f"{part.capitalize()} {PART}% ({'fatty' if part == 'bottom' else 'water-rich
           xlabel="q (nm ⁻¹)", xlim=(2, 24), ylim=(-0.05, 1.3))
    ax.set_xticks([5, 10, 15, 20])
    ax.grid(alpha=0.3)
    ax.legend(frameon=False, loc="upper left", fontsize=6)
axes[0].set_ylabel("I(q), prepared (fat peak = 1)")
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

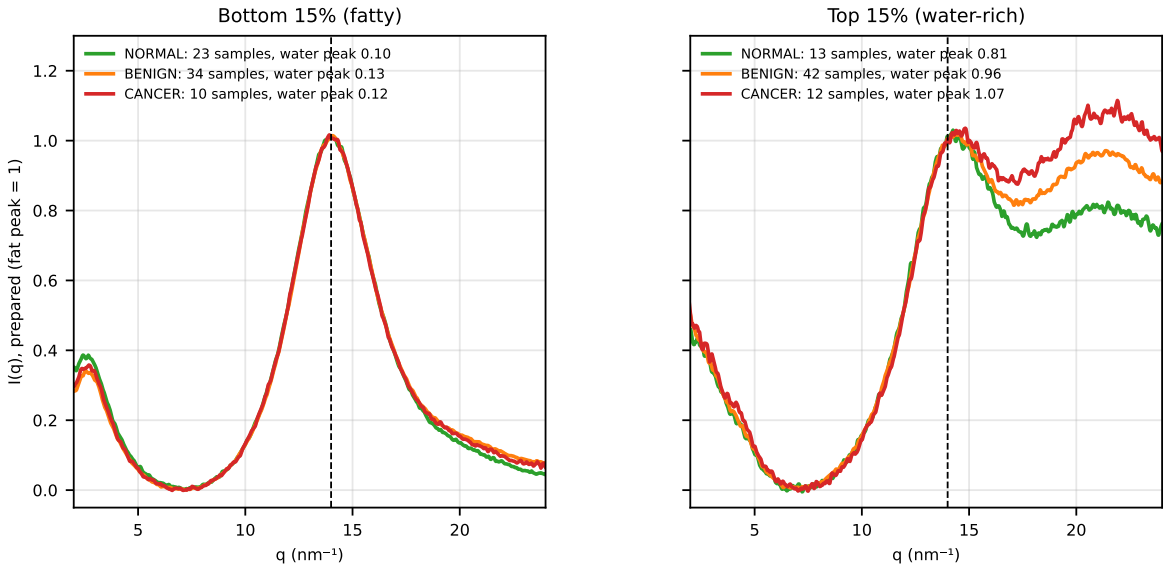


Figure 1.2: Mean prepared curve of each cohort's samples in the fattiest bottom 15% (**left**) and the wettest top 15% (**right**) of Figure 1.1. The three cohorts overlap when fatty; they separate as NORMAL < BENIGN < CANCER in the wettest 15%.

2 Our data match synchrotron and ex-vivo data

> **Finding 2**
>
> The wettest 15% of our samples order as NORMAL < BENIGN < CANCER (Figure 1.1, Figure 1.2), matching the synchrotron and Keele ex-vivo ordering of less fat and more water towards disease (Figure 2.1, Figure 2.2). Under the fixed water-peak threshold of Appendix C, BENIGN and CANCER look almost the same and the paired test only supports **CANCER vs NORMAL**; BENIGN vs NORMAL is 10/27 (37%) and CANCER vs BENIGN 2/5 (40%), neither above chance (Appendix D).

2.1 Synchrotron data

Figure 2.1 shows the molecular form factors of excised breast tissue measured with synchrotron radiation by Conceição, Poletti and colleagues [10] [11]; the dataset was supplied by A. L. C. Conceição. Their discriminant analysis found the ratio of the peaks at q = 19.8 and 13.9 nm ^{1}, the water and fat peaks, to be one of the features that best separates the tissue types [11], as earlier synchrotron work had also found [12]. The normal tissues (peripheral normal, adipose and glandular) are dominated by the fat peak at q 14 nm ^{1} and fall away above q 17. Malignant tissue has a much weaker fat peak and a broad water peak around q 19-20, and benign tissue lies between the two. Tissue with less fat and more water has a higher water peak relative to its fat peak.

The right panel prepares the synchrotron curves **exactly as our scans**: the same peak-shift correction (fat peak moved to q = 14 nm ^{1}) and the same min-max normalisation (Appendix A). The comparison with Figure 1.2 is therefore like for like.

```
import zipfile
import xml.etree.ElementTree as ET

ANDRE = DATA_DIR / "Andre_FormFactor_dataset.xlsx"  # sheet 1: q, then one column per sample
ns = {"m": "http://schemas.openxmlformats.org/spreadsheetml/2006/main"}

with zipfile.ZipFile(ANDRE) as zf:
    shared = ["".join(t.text or "" for t in si.findall(".//m:t", ns))
```

```
            for si in ET.fromstring(zf.read("xl/sharedStrings.xml")).findall("m:si", ns)]

    def cell_val(c):
        v = c.find("m:v", ns)
        if v is None or v.text is None:
            return None
        if c.attrib.get("t") == "s":
            return shared[int(v.text)]
        try:
            return float(v.text)
        except ValueError:
            return v.text

    def load_sheet(name):
        sheet = ET.fromstring(zf.read(f"xl/worksheets/{name}"))
        return [{"".join(ch for ch in c.attrib.get("r", "") if ch.isalpha()): cell_val(c) for
                 for row in sheet.findall("m:sheetData/m:row", ns)]

    sheet1 = load_sheet("sheet1.xml")
    sheet2 = load_sheet("sheet2.xml")

meta = {}
for row in sheet2[1:]:
    sid = str(row.get("A", "")).strip().replace(" ", "")
    if sid:
        meta[sid] = (str(row.get("B", "")).strip(), str(row.get("C", "")).strip().lower())

col_to_sample = {col: str(name).replace(" ", "") for col, name in sheet1[0].items() if col !=
andre_q = []
series = {col: [] for col in col_to_sample}
for row in sheet1[1:]:
    if row.get("A") is None:
        continue
    andre_q.append(float(row["A"]))
    for col in col_to_sample:
        v = row.get(col)
        series[col].append(np.nan if v is None else float(v))
andre_q = np.asarray(andre_q)

groups = {"peripheral normal": [], "normal adipose": [], "normal glandular": [], "malignant"
for col, sid in col_to_sample.items():
    if sid not in meta:
```

```
        continue
    lab, cls = meta[sid]
    y = np.asarray(series[col], dtype=float)
    if lab == "peripheral_normal_tissue":
        groups["peripheral normal"].append(y)
    elif lab == "normal_adipose":
        groups["normal adipose"].append(y)
    elif lab == "normal_glandular":
        groups["normal glandular"].append(y)
    elif cls == "malignant":
        groups["malignant"].append(y)
    elif cls == "benign":
        groups["benign"].append(y)

andre_colours = {"peripheral normal": "#2c7bb6", "normal adipose": "#1a9641", "normal glandul
                 "malignant": "#d7191c", "benign": "#fdae61"}

# the same preparation as our scans: fit the fat peak (Gaussian + straight line over q 10-17
import nova_sdd_peak_align as align

def prepared(curve):
    """The curve on andre_q with its fat peak moved to q = 14 and stretched (0 at the trough
    fit = align.fit_peak(andre_q, curve)
    if fit is None or not (align.SIGMA_RANGE[0] <= fit["sigma"] <= align.SIGMA_RANGE[1]) \
            or not (align.FAT_WINDOW[0] < fit["peak"] < align.FAT_WINDOW[1]):
        return None
    shifted_q = andre_q * align.TARGET_Q / fit["peak"]
    refit = align.fit_peak(shifted_q, curve)
    if refit is None:
        return None
    stretched = ne.normalised_curve(shifted_q, curve, refit["value"], "stretch")
    return None if stretched is None else np.interp(andre_q, shifted_q, stretched, left=np.na

prepared_groups = {name: [p for p in map(prepared, curves) if p is not None] for name, curves

plt.rcParams["font.size"] = 7  # small text to match the other figures at A4 width
fig, (raw_ax, prep_ax) = plt.subplots(1, 2, figsize=(8, 3.8))
for ax, these in ((raw_ax, groups), (prep_ax, prepared_groups)):
    for name, curves in these.items():
        mat = np.vstack(curves)
        mean, std = np.nanmean(mat, axis=0), np.nanstd(mat, axis=0)
        left_out = len(groups[name]) - len(curves)
```

```
        note = f", {left_out} without a clear fat peak left out" if left_out else ""
        ax.plot(andre_q, mean, lw=1.8, color=andre_colours[name], label=f"{name} (n = {len(cu
        ax.fill_between(andre_q, mean - std, mean + std, color=andre_colours[name], alpha=0.1
    ax.set(xlabel="q (nm ⁻¹)", xlim=(0, 25))
    ax.set_ylim(bottom=0)
    ax.grid(alpha=0.3)
    ax.legend(frameon=False, fontsize=6)
raw_ax.set(ylabel="Intensity (a.u.)", title="As measured")
prep_ax.axvline(14, color="black", ls="--", lw=0.8)
prep_ax.set(ylabel="I(q), stretched (fat peak = 1)", ylim=(-0.05, 1.6), title="Prepared like
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

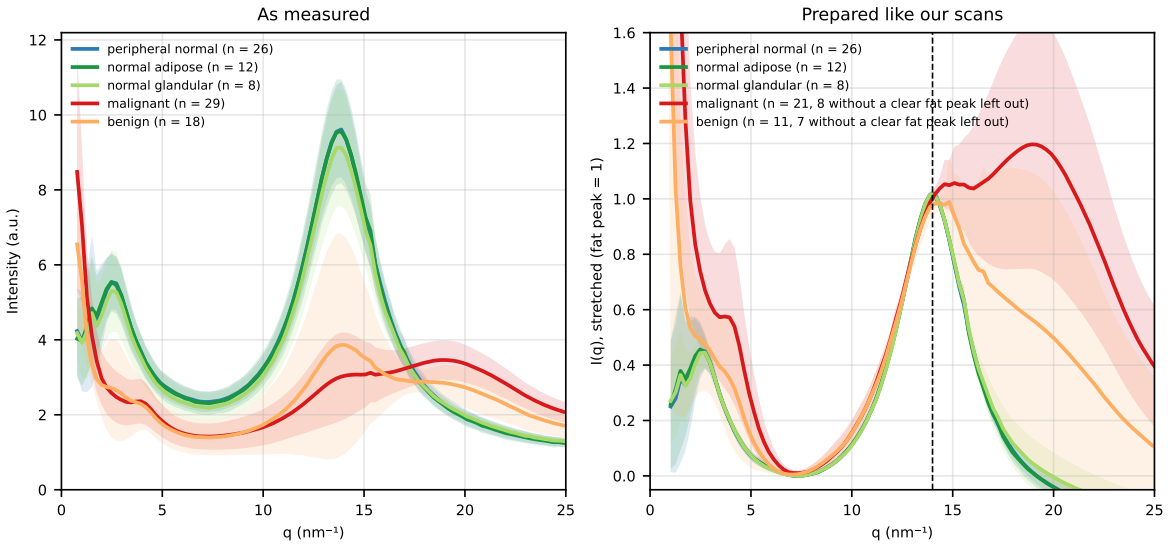


Figure 2.1: Synchrotron form factors of breast tissue [10, 11]: the mean of each tissue type, with ±1 standard deviation shaded. **Left**: as measured. **Right**: prepared exactly like our scans (fat peak moved to q = 14, then 0 at the trough and 1 at the fat peak), for direct comparison with Figure 1.2; curves without a clear fat peak cannot be prepared this way and are left out.

2.2 Keele ex-vivo data

Figure 2.2 shows ex-vivo scans of breast tissue measured at Keele University [13], one heatmap each for Healthy, Benign and grade 3 cancer tissue, each row one normalised scan.

The water peak is missing from these scans: they stop at q 17 nm ^{1}, below the water peak at q 20-22, so we cannot see the water rise directly. **The reduction in the fat peak, however, is clear**: in the Healthy tissue a large share of the scans are dominated by a bright fat peak at q 14, in Benign it covers fewer, and in the grade 3 cancer it covers a smaller share still. Less fat is the other side of the same change that our water-rich scans and the synchrotron data show as a higher water peak.

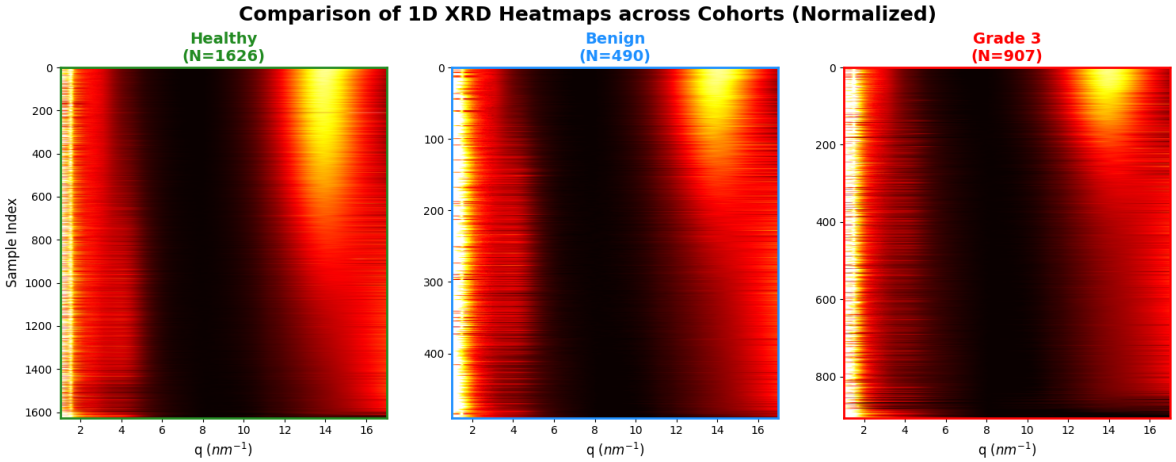


Figure 2.2: Ex-vivo XRD scans of breast tissue measured at Keele, normalised, one heatmap per cohort: Healthy, Benign and grade 3 cancer. The scans stop at q 17 nm ^{1}, so the water peak is not measured, but the fat peak at q 14 clearly weakens towards cancer.

3 Rigorous targeting of the tumour is not necessary

Finding 3

Rigorous targeting of the tumour is not necessary. The tumours we have analysed are small — 2–18 mm on biopsy (62 breasts) and 5–95 mm on imaging (69 breasts) — and each breast is measured with only about 3 beams, placed approximately from the mammogram or ultrasound. The chance that those beams hit the tumour itself is small. Yet where the beams go through glandular, water-rich tissue, the CANCER breast has the higher Water:Fat in **15/18 (83%)** of patients (Appendix C). The data point to **wide-reaching changes in the breast, but in the glandular tissue**, not to a signal that needs the tumour in the beam.

3.1 The tumours are small; the beams are few

```
plt.rcParams["font.size"] = 7
fig, (biopsy_ax, imaging_ax) = plt.subplots(1, 2, figsize=(8, 3.6), sharey=False)
rng = np.random.default_rng(0)

def plot_usable_rejected(ax, usable_vals, rejected_vals, ylabel, title):
    groups = [(0, usable_vals, TOP_COLOUR, "usable scan"), (1, rejected_vals, "grey", "reject
    box_with_points(ax, [(x, list(vals), colour) for x, vals, colour, _ in groups if len(vals
    ax.set_xticks([0, 1], [f"usable\n(n = {len(usable_vals)})", f"rejected only\n(n = {len(re
    ax.set(ylabel=ylabel, title=title, ylim=(0, None))
    ax.grid(alpha=0.3, axis="y")
    from matplotlib.patches import Patch
    ax.legend(handles=[Patch(color=TOP_COLOUR, alpha=0.5, label="usable (water-rich) scan"),
                       Patch(color="grey", alpha=0.5, label="rejected scans only")],
              frameon=False, loc="upper right", fontsize=6)

plot_usable_rejected(biopsy_ax, biopsy_use, biopsy_rej, "Linear extent (mm)", "Biopsy cores")
plot_usable_rejected(imaging_ax, imaging_use, imaging_rej, "Largest mass on imaging (mm)", "I
```

```
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

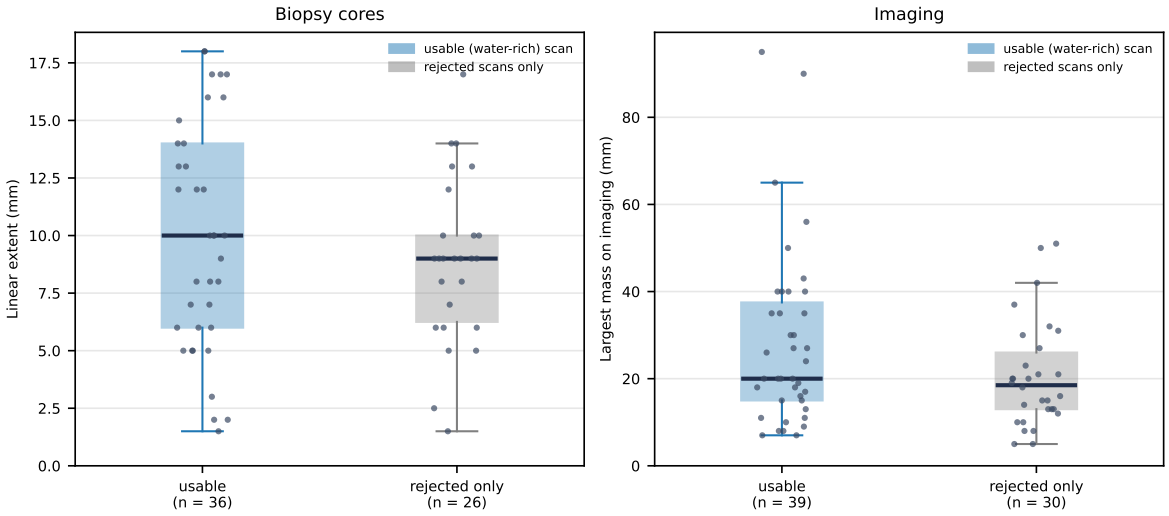


Figure 3.1: Tumour size of every CANCER breast with a reported size. **Left**: linear extent in the biopsy cores (pathology). **Right**: largest mass on imaging (mammogram, ultrasound or MRI). Blue: breast has at least one usable, water-rich scan; grey: only rejected (fatty) scans. One dot per breast.

Over every CANCER breast with a reported size, the tumours run from 2 to 18 mm on biopsy (median 9 mm, 62 breasts) and from 5 to 95 mm on imaging (median 20 mm, 69 breasts). Usable and rejected-only breasts have similar size ranges (imaging medians 20 mm and 18 mm), so having a water-rich scan is not simply a matter of a larger tumour. The biopsy size is only the length of tumour along a needle core; imaging measures the mass itself and is the better sense of how large a target the tumour is. Even so, most are a couple of centimetres or less. Size by grade is in Appendix H.

Each breast is scanned only a few times — typically 3 beams — and the operator places them from a quadrant, a clock position and a distance from the nipple. The breast sits differently on our instrument than on the mammogram. Hitting a 20 mm mass with 3 approximately placed beams is unlikely to be the rule; missing the tumour altogether will often be the case. We have no data on where each beam landed relative to the tumour, so this is an argument from the sizes and the scanning practice, not a measured miss rate.

3.2 Yet the glandular signal holds

What we do measure is whether each beam went through **water-rich, glandular tissue** (Chapter 1, Appendix C). When both breasts have at least one such scan, the CANCER breast is the wetter one in most CANCER/NORMAL patients, across the size range of Figure 3.1. Fatty scans of the same breasts do not carry the signal. So the result does not require the tumour in the beam; it requires glandular tissue.

A working **hypothesis** is that the cancer changes tissue beyond the tumour itself — a **sphere of influence** — and that those changes are expressed in the glandular/fibrous tissue of the breast. Tumours remodel the fibrous stroma around them [5], and X-ray scattering has detected structural changes in and next to breast tumours [6] [7]. How far that region reaches cannot be measured from these data. What the data support is the practical claim: **high positioning accuracy on the tumour is not needed**; what matters is that each scan reaches glandular tissue.

4 A proposal for Human 2.0: mapping the breast

> **Finding 4: a proposal for Human 2.0**
>
> This chapter is a proposal, not a result. Findings 1-3 say a measurement has to reach water-rich tissue. On Human 1.0 the water-peak rule (Appendix C) could do this during scanning, as a stopgap. We recommend **mapping** the breast for Human 2.0: it applies the same water-peak idea **after** the scan, over the whole breast.

4.1 A Human 1.0 stopgap: the water-peak filter during scanning

The water-peak rule of Appendix C could be applied while the patient is on the instrument. At each position it needs a short scan, long enough to see the fat and water peaks; a quick analysis of that scan (correct the peak position, normalise, and check the water peak height against 0.35 with the water centred at q 19); and a decision: if the scan is water-rich the full scan proceeds, and if not the operator moves to a clearly different spot. The scanning software would need this analysis built in, much like the existing transmission check (Phase 1) that decides whether a position is scanned. In theory, this could be implemented on Human 1.0. It still depends on where the operator puts the beam, and gives a few points per breast.

4.2 The recommendation: mapping

To **map** the breast is to scan a grid of points, each for a short time, and decide afterwards which points to use. Each point gives a full 1D curve I(q), so a map is a **3D matrix**: two positions (x, y) and q. Each point holds a curve, and each q holds an image of the breast at that q. Figure 4.1 shows an example measured at Ulster University on a tissue sample: a 26 × 25 grid at 0.5 mm spacing, 1 s per point, with 336 points measured and 210 of them on the tissue sample. The rest see only the empty holder. No background is subtracted.

Mapping works much like an **XRD video rather than a snapshot**. At every point we gather a frame. Each frame on its own has a low XRD signal-to-noise ratio, but it can be used as it is for transmission, giving high spatial resolution. Conversely, we can sum over more frames,

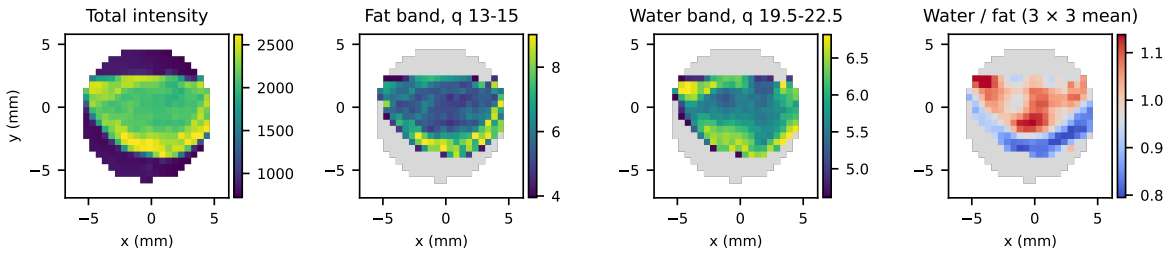


Figure 4.1: The Ulster map, one value per point. **Left to right**: the total scattered intensity (the sample against the empty holder), the fat band (q 13-15), the water band (q 19.5-22.5), and the water band over the fat band, with each point averaged with its neighbours (3 × 3). Grey points are off the sample; white points were not measured.

covering more distance, for a better XRD signal-to-noise ratio at lower spatial resolution. The trade-off is chosen after the scan, and can be different for different questions.

A single 1 s point gives a noisy curve, but neighbouring points can be added together. Figure 4.2 shows one point in the middle of the sample on its own, and the mean of its 3 × 3 and 5 × 5 neighbourhoods and of every point on the sample: the fat and water peaks only become clear once points are added. On their own, the points are as noisy as the differences across the sample: neighbouring points differ in water/fat by about **0.10** from noise alone, while the 3 × 3 means range from 0.82 to 1.12 across the sample. Averaging n points cuts the noise by about $\sqrt{n}$, by 3 for a 3 × 3 block.

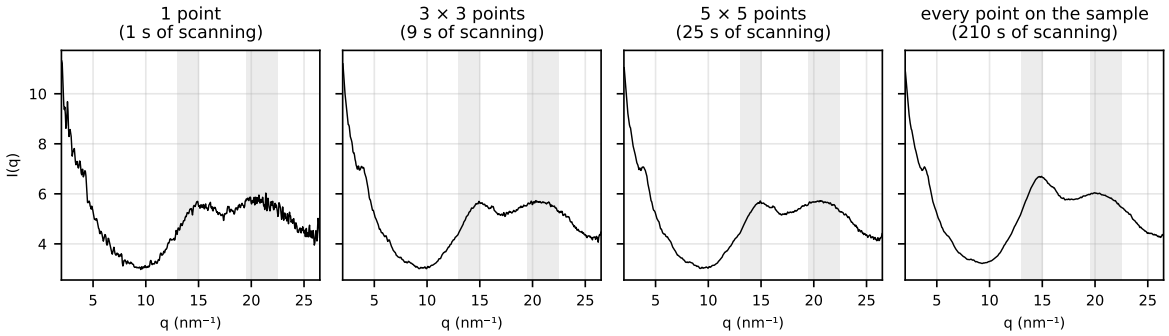


Figure 4.2: Adding points improves the 1D curve. One point in the middle of the sample on its own, and the mean of its 3 × 3 and 5 × 5 neighbourhoods and of every point on the sample. The grey bands mark the fat and water bands.

Because every point is kept, the map can be used in two ways **after** the scan:

1. **Select regions.** Points can be grouped by what they show — in particular the **wettest 15%** within each breast (Figure 1.1, Figure 1.2) — and only those used for the breast's Water:Fat. That is a percentile inside the breast, not a fixed threshold on a few single

points. A fixed water-peak threshold can still drop empty or too-dry points first; the ranking then picks the best tissue that remains. Figure 4.3 splits the sample into its most fatty, middle and most water-rich thirds by the 3 × 3 water/fat: the mean curves differ clearly, the fattiest with a fat peak standing above the water and the most water-rich with the two about level.

2. **Compare whole breasts.** The distribution of water/fat over every point of one breast can be compared with that of the other breast, rather than a few single points. This uses the whole breast and does not depend on where the tumour is (Chapter 3).

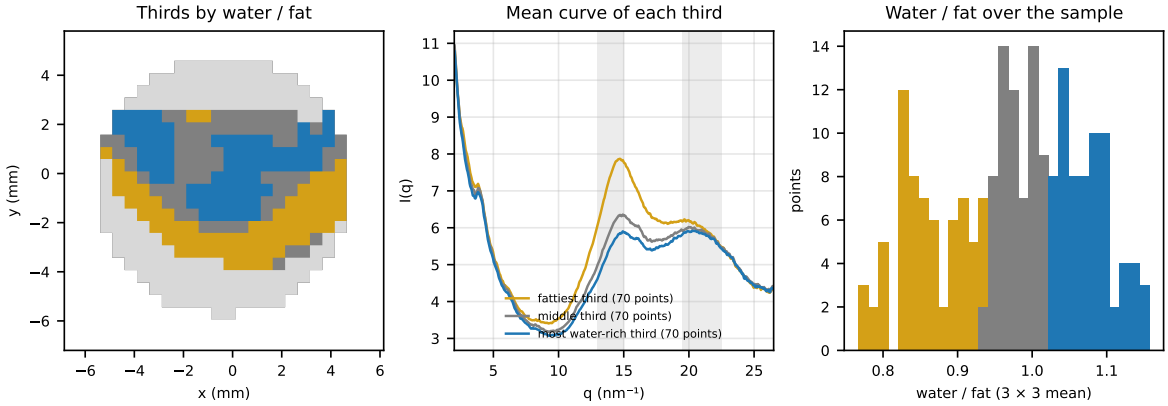


Figure 4.3: Selecting regions after the scan. **Left**: the sample split into thirds by the 3 × 3 water/fat (fattiest, middle, most water-rich). **Middle**: the mean curve of each third. **Right**: the distribution of the 3 × 3 water/fat over every point of the sample, the kind of whole-breast summary that could be compared between a patient's two breasts.

Mapping gives a **whole-breast analysis**: we can tell the adipose and glandular areas apart, analyse them separately if we want, and get an idea of **where** interesting tissue is — on the hypothesis of a tumour sphere of influence (Chapter 3) — without aiming at the tumour or deciding at the instrument.

4.3 What a whole-breast map would let us measure

With about three points per breast we get one noisy Water:Fat number, so we can only ask whether the mean is higher in the CANCER breast. A map of around 100 points per breast gives a **distribution** per breast. That supports several readouts (proposal only, not results):

- **Wettest region.** A high quantile, or the mean of the wettest N points (not the single maximum), compared left against right.

- **Fattiest region as baseline.** Finding 1 says the fattiest 15% look the same across cohorts, so this is a within-patient control, not a discriminator: thickness, calibration and density offsets cancel. Wettest minus fattiest (or their ratio) is a natural per-patient measure.
- **Mean Water:Fat** per breast, left against right (paired, so it controls for the patient).
- **Variance of Water:Fat** per breast, left against right. Correct for counting noise from neighbour-point differences (about 0.10 on the Ulster map), and report a coefficient of variation or work on the log ratio, because variance scales with the mean. Mask edge and thin points.
- **Comparison to a "normal" reference** (mean or variance) is deferred: it needs a healthy-volunteer cohort, and normal varies with density, age and menopausal status.

A map can separate a **global** effect (the mean shifts), a **local** effect (the wet tail grows) and a **heterogeneity** effect (the variance changes). Single-point scans cannot. It also gives internal controls, several independent readouts, and spatial information on where the wet tissue is, which ties to the tumour sphere-of-influence idea in Chapter 3 (Chapter 3).

For a small pilot, pre-specify one **primary** endpoint (wettest region, left against right) and treat the rest as secondary or exploratory, to avoid multiple-comparison problems. Points within a breast are spatially correlated, so analyse at breast or patient level, not as 100 independent samples. Prefer comparing whole distributions (for example several quantiles) over relying on a single summary number.

4.4 How long would a whole-breast map take?

The practical picture is a **continuous fly-scan**, not a stop-and-dwell grid. The detector runs at **1 Hz**, and the source and detector move together at a preset speed. That speed sets how long the beam spends on each volume of the breast, and so how many photons each 1 s frame collects. Dose and XRD signal-to-noise both follow from that choice.

Surface scan first (radiation off). A very rapid initial pass, under 10 s, measures the thickness of the breast at each point with the beam off. That surface map does three jobs:

1. **Choose where to scan.** Regions below a thickness cut (for example thinner than 2 cm) can be left out, so time and dose go only where the breast is thick enough to be useful.
2. **Support q corrections.** Thickness (or distance) can be used to correct the curves if needed. The Water:Fat ratio analysis itself does not require a distance measurement.
3. **Turn transmission into linear attenuation.** With thickness known, transmission becomes a linear attenuation map — a measure of breast density — at **high spatial resolution**. That transmission map is valuable on its own, and it is also what later marks regions of interest for XRD analysis.

Then the XRD scan. Once the region of interest is set from the surface and transmission maps, a scan protocol is designed to keep the time down. Speed can be **dynamic**: slower over thicker tissue so enough photons are collected, faster where the breast is thinner. Neighbouring 1 Hz frames can still be summed afterwards for a cleaner XRD curve at the cost of spatial resolution (Figure 4.2); the transmission values keep the fine grid.

We expect these scans to take **no longer than 5 minutes**. A simulated half breast, stepped at 5 mm only where it is at least 2 cm thick, is 166 such seconds — 2 min 46 s (Chapter 5). Absolute dose in mGy has not been estimated here and has to be confirmed on the instrument.

> **i** Mapping is the recommendation for Human 2.0
>
> Mapping gives a whole-breast analysis: noisy single points are summed into clean curves, and which regions to analyse is decided **after** the scan. The water-peak rule (Appendix C) is a possible stopgap on Human 1.0.

5 Simulating a Human 2.0 map

> **i** A worked example of the map, not a patient result
>
> The first map is a breast with a **2 cm** tumour. Two million photons is what the real instrument collects in **1 second**, so each step of the map is 1 s of beam time. This breast was stepped at **5 mm**, and only where it is **2 cm or thicker**: **166 steps**, which is **166 seconds (2 min 46 s)**. A dense breast with a **2 cm** tumour follows. The comparison with each breast once the tumour is removed is at the end.

The simulator builds a 3D breast from adipose and a branching glandular network, and can add a tumour of any chosen size. The volume used here is a **half breast**: semi-axes 5.5 cm in x and y, height 5.5 cm, voxels of 5 mm, on a 24 × 12 × 12 grid at 0.5 cm (Figure 5.1). Materials are assigned per voxel (adipose, then glandular).

A run has two stages. First the photons interact in the breast and are stored. Then they are passed to a detector. The detector and the sample-to-detector distance can be changed after the breast has been simulated, without repeating it. The detector here is the **CdTe Dectris 500K** that Human 2.0 will move to.

Figure 5.2 is one such run, 2 million photons, aimed at a **very fatty** part of this breast. About 16% of the beam is transmitted, and the 1D curve has the fat peak near q = 14 nm ^{1} and little water, the same shape as the fatty scans in the real data.

5.1 A map of every thick point

The same 2-million-photon run was repeated on the UCL cluster at every point of this breast that is 2 cm or thicker. That is the scan proposed for Human 2.0 (Chapter 4): leave out the thin edge, and give each remaining point one second. At 5 mm steps that is 166 points, so **2 min 46 s** of beam time, inside the 5-minute limit.

This map is a breast with a **2 cm** tumour (Figure 5.3). Each simulated point was prepared exactly as a real scan: the fat peak was shifted to q = 14 nm ^{1}, and the curve was normalised the same way. Thickness and transmission give the linear attenuation, $= -\ln(T) / \text{thickness}$. A 3 × 3 kernel, nine neighbouring points, was slid over every position, and the search keeps the highest Water:Fat automatically.

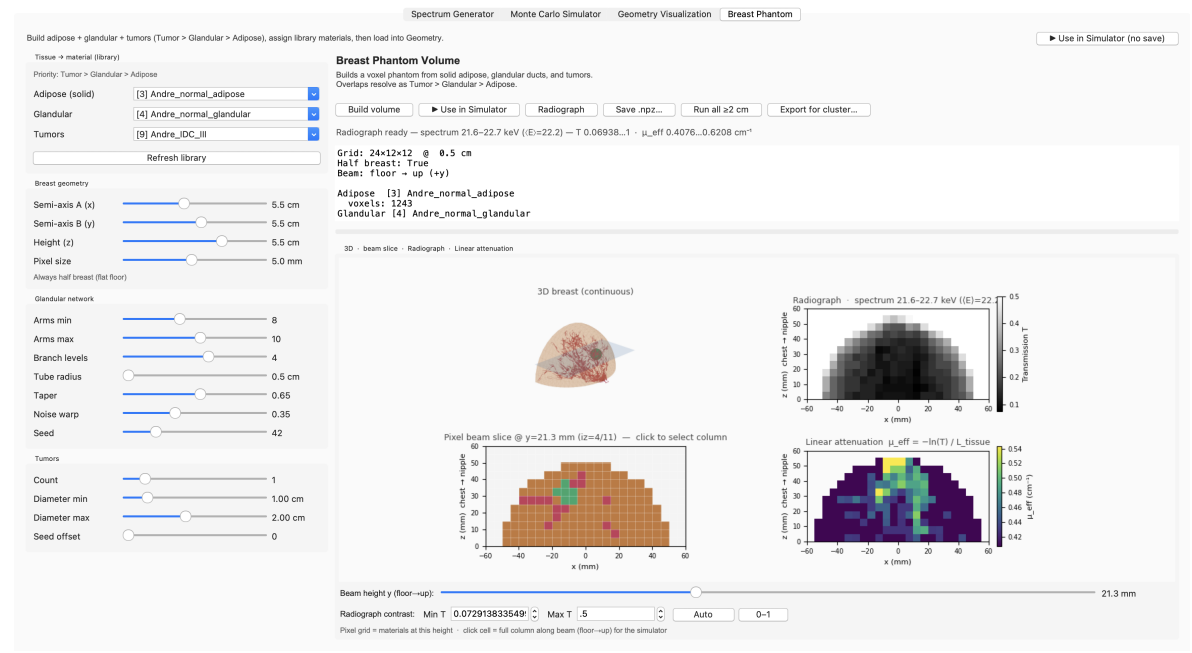


Figure 5.1: The phantom generator. Left: the breast size (5.5 cm semi-axes and height, 5 mm voxels) and the glandular network. Right: the half-breast volume, its radiograph, a beam-height slice, and the linear attenuation from that radiograph.

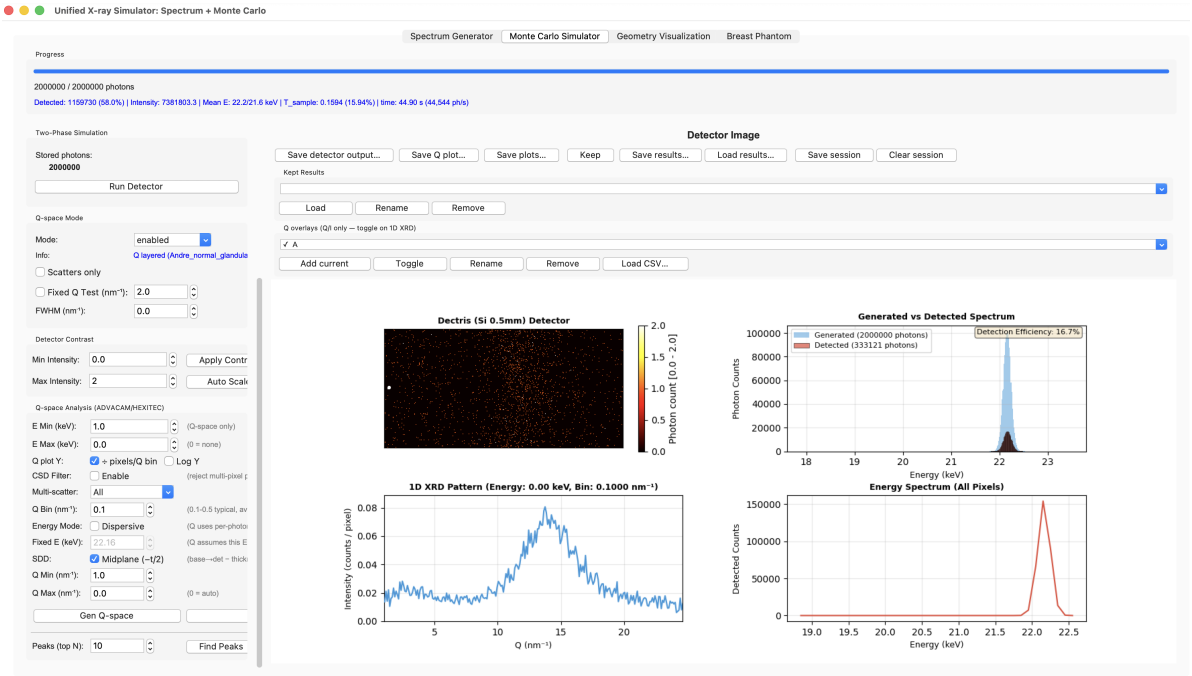


Figure 5.2: One 2-million-photon run on the CdTe Dectris 500K, after the stored photons have been passed through the detector. The 1D curve is the fatty point used as the example.

Figure 5.4 puts the two readings of this breast side by side. On the left, linear attenuation behaves like a mammogram: glandular tissue and the tumour are both dense, so both are bright. On the right, the 3×3 kernel map stays low over the rest of the breast, and the red box is the kernel the search kept. It sits on the tumour.

5.2 A dense breast, with a 2 cm tumour

Figure 5.5 is a dense breast: glandular tissue fills much more of the slice, and a **2 cm** tumour sits in it. Figure 5.6 is the same pair of readings as Figure 5.4. With more glandular tissue, linear attenuation shows one broad dense region, so the tumour is harder to separate from the tissue around it. The 3×3 search still outlines it. The red box is that kernel, Water:Fat **0.337**, on the tumour.

5.3 Compared with no tumour

The same 3×3 search, on the same breast with the tumour removed and nothing else changed. With a tumour the summed curve rises through the water band. Without one it stays with the global mean, and the breast would fall on the fatty side of the top 15% cut (Chapter 1) and of the water-peak rule at 0.35 (Appendix C, Figure F.5).

5.3.1 The less dense breast

Figure 5.7 is the less dense breast with the 2 cm tumour. The kernel the search keeps has Water:Fat **0.292**. Figure 5.8 is that breast with the tumour removed. The wettest kernel is then **0.176**, and the red curve lies on the global mean through the water band.

5.3.2 The dense breast

Figure 5.9 is the dense breast with the 2 cm tumour. The kernel the search keeps has Water:Fat **0.337**. Figure 5.10 is that breast with the tumour removed. The wettest kernel is then **0.192**, and the red curve stays with the global mean.

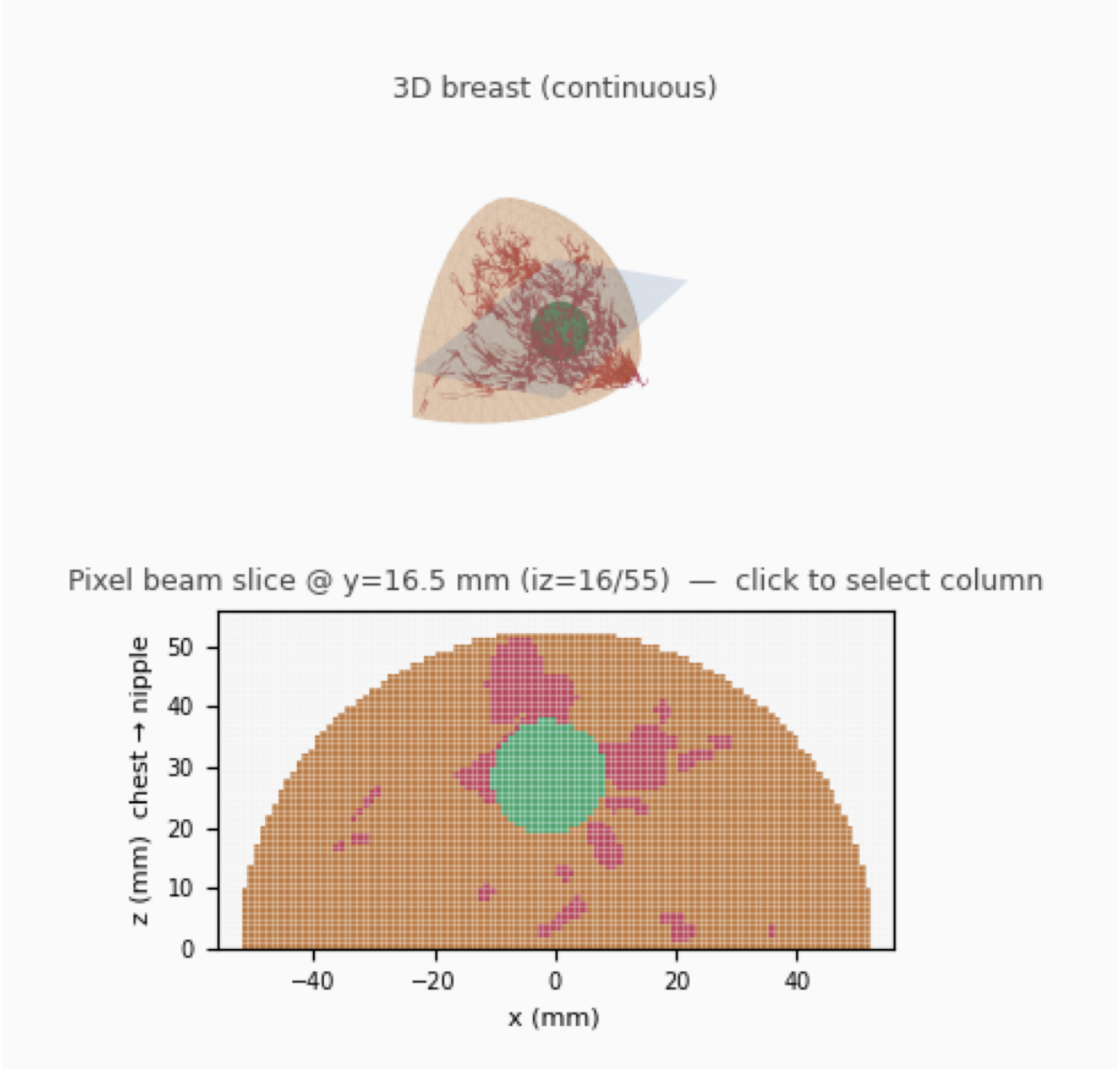
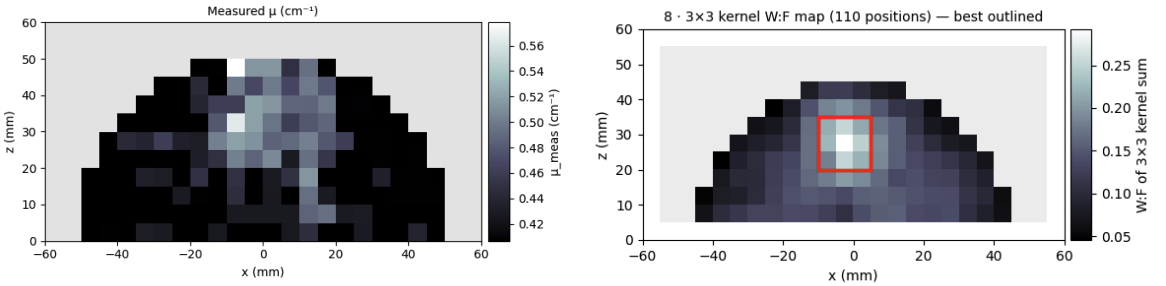


Figure 5.3: A breast with a 2 cm tumour. Top: the volume, with the tumour circled. Bottom: one beam slice; the tumour is the green mass, glandular tissue is pink, adipose is brown.



(a) Linear attenuation, = −ln(T) / thickness.

(b) Water:Fat of every 3 × 3 kernel (110 positions). The red box is the wettest.

Figure 5.4: Linear attenuation and the 3 × 3 kernel map of the breast with the 2 cm tumour. Left: dense tissue, glandular and tumour together, as on a mammogram. Right: the kernel the search kept, outlined, on the tumour.

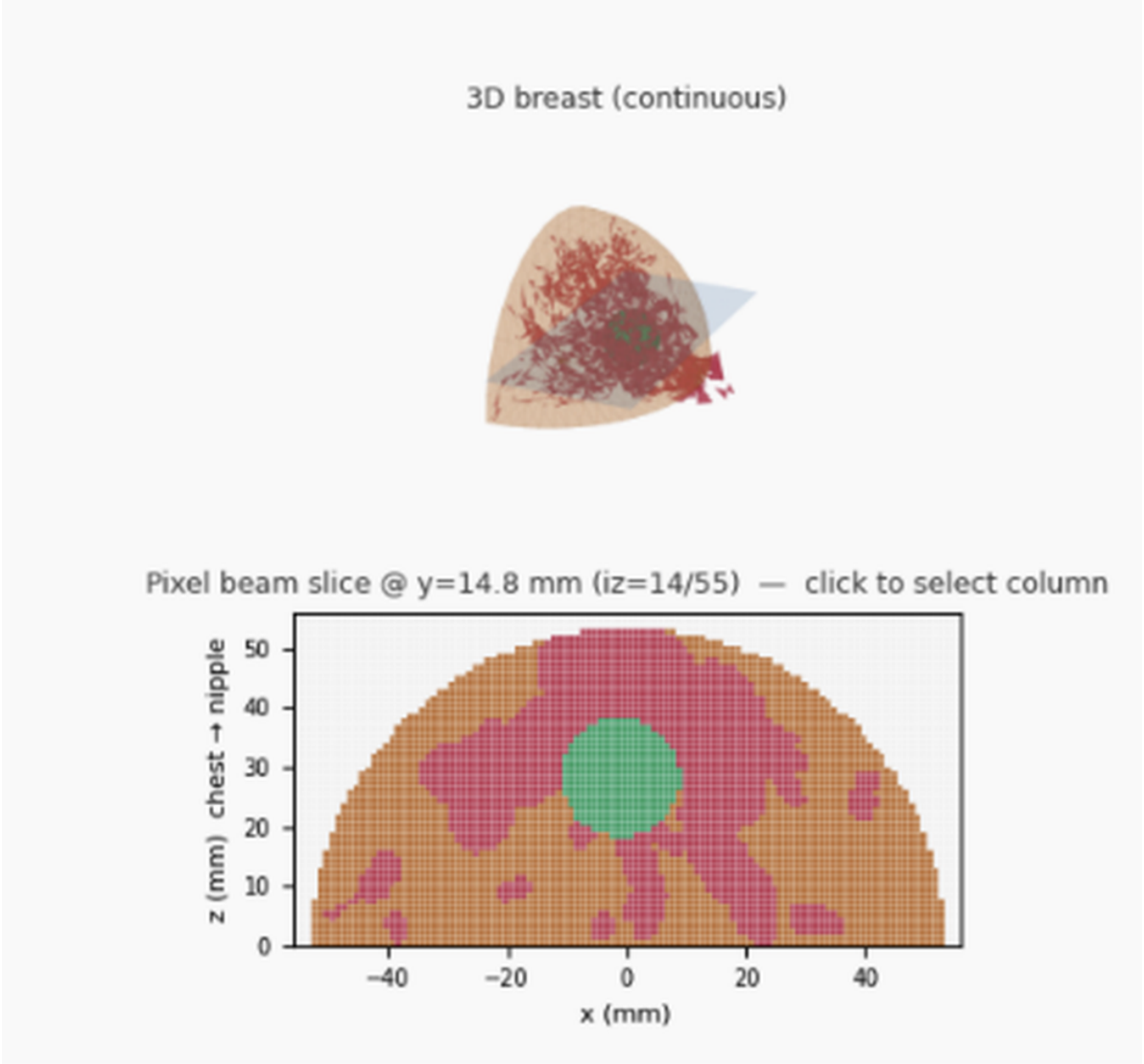
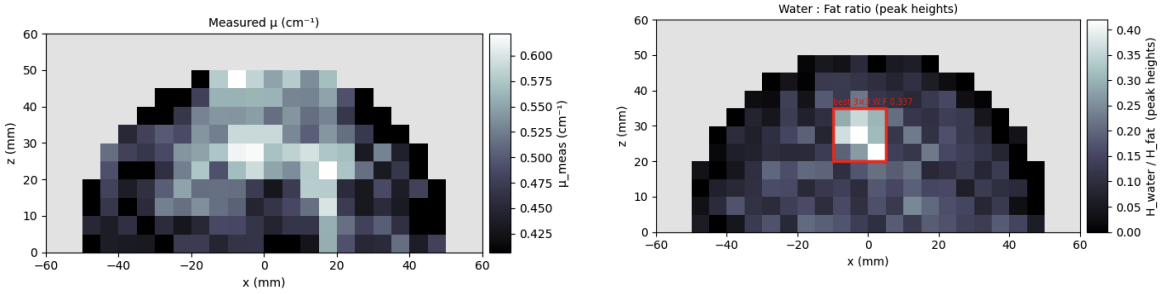


Figure 5.5: A dense breast with a 2 cm tumour. Top: the volume, with the tumour circled. Bottom: one beam slice; the tumour is the green mass, glandular tissue is pink, adipose is brown.



(a) Linear attenuation, $= -\ln(T) / \text{thickness}$.

(b) Water:Fat from the peak heights. The red box is the wettest 3 × 3 kernel.

Figure 5.6: Linear attenuation and the 3 × 3 kernel map of the dense breast with a 2 cm tumour. Left: glandular tissue and the tumour form one dense region. Right: the kernel the search kept, Water:Fat 0.337, on the tumour.

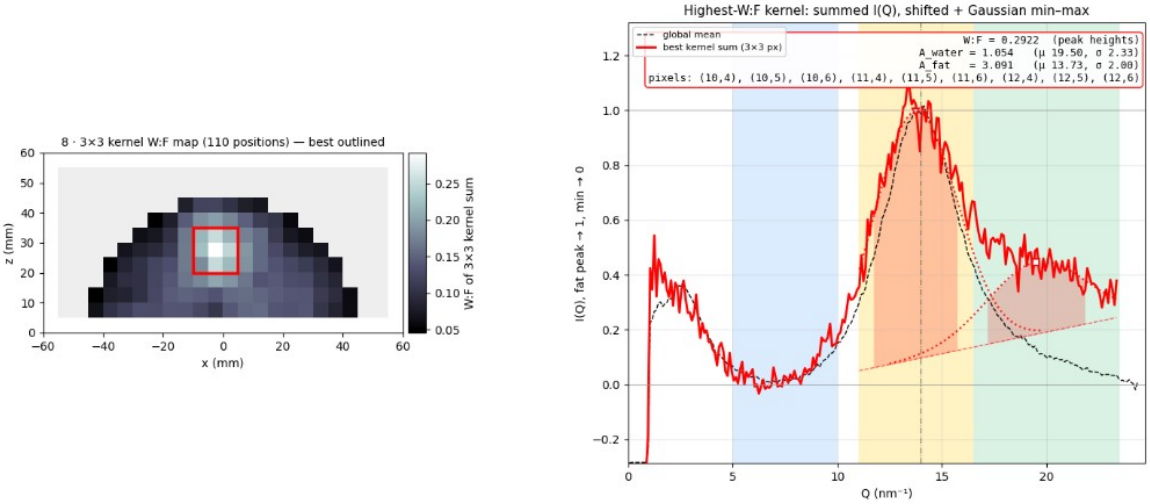


Figure 5.7: The wettest 3 × 3 kernel on the less dense breast with a 2 cm tumour. Left: Water:Fat of every 3 × 3 position (110 of them); the search outlines the wettest. Right: the summed curve (red) against the global mean (dashed). Water:Fat of the kernel is 0.292.

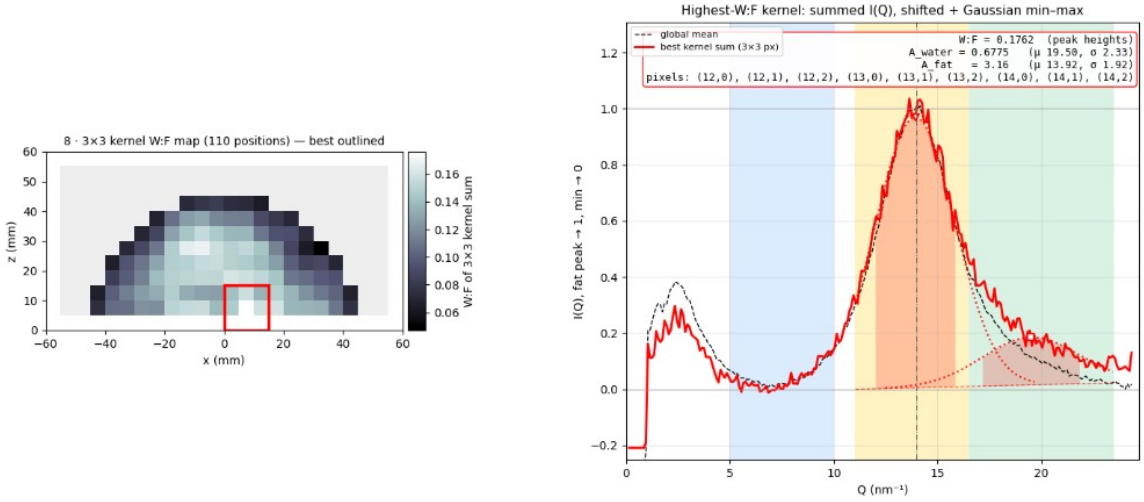


Figure 5.8: The wettest 3 × 3 kernel on the less dense breast with no tumour. Left: Water:Fat of every 3 × 3 position (110 of them); the search outlines the wettest. Right: the summed curve (red) against the global mean (dashed). Water:Fat of the kernel is 0.176.

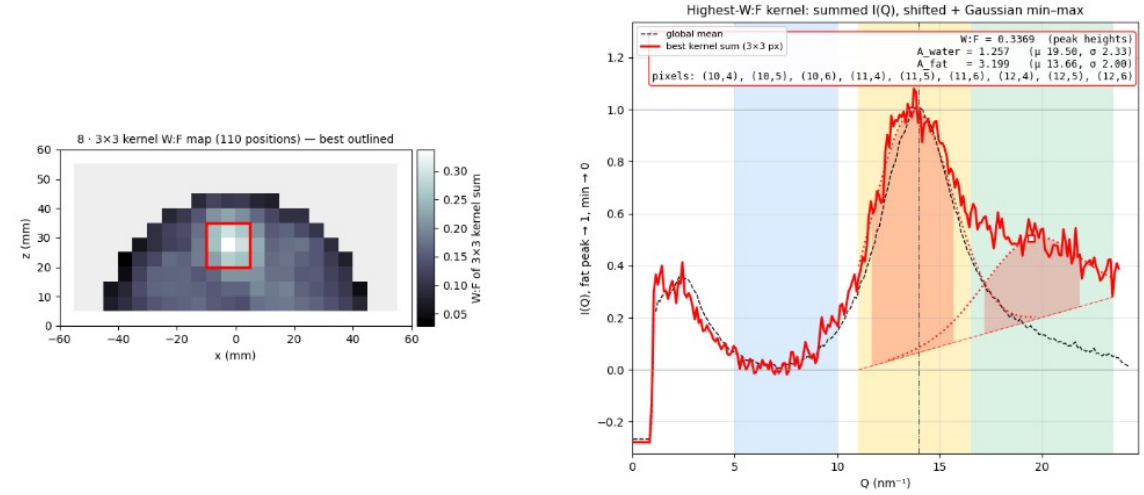


Figure 5.9: The wettest 3 × 3 kernel on the dense breast with a 2 cm tumour. Left: Water:Fat of every 3 × 3 position (110 of them); the search outlines the wettest. Right: the summed curve (red) against the global mean (dashed). Water:Fat of the kernel is 0.337.

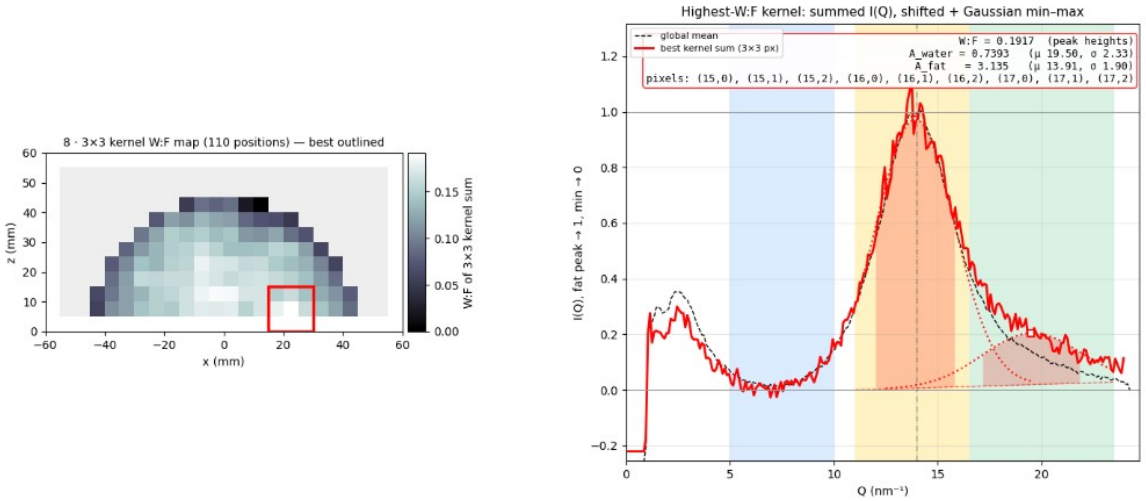


Figure 5.10: The wettest 3 × 3 kernel on the dense breast with no tumour. Left: Water:Fat of every 3 × 3 position (110 of them); the search outlines the wettest. Right: the summed curve (red) against the global mean (dashed). Water:Fat of the kernel is 0.192.

6 Limitations

⚠ What limits these results

1. **Small numbers.** The headline rests on 18 CANCER/NORMAL patients: 83%, with a 95% confidence interval of 59%–96%. It is a pilot result.
2. **Only CANCER vs NORMAL is supported on the threshold rule.** The wettest 15% of samples separate NORMAL < BENIGN < CANCER (Figure 1.2), but under the fixed usable-scan cut of Appendix C BENIGN and CANCER look almost the same, and the paired test for BENIGN vs NORMAL is 10/27 (37%) and for CANCER vs BENIGN 2/5 (40%), neither above chance. Distinguishing benign from cancer is what clinicians need most, and this analysis does not yet do it. **Mapping would address this:** a whole-breast map lets us keep the wettest tissue in each breast (the top-15% idea of Chapter 1) and compare that, instead of a few single points that only just clear a fixed threshold (Chapter 4).
3. **Few young CANCER breasts.** Only 2 CANCER breasts under 40 have a usable scan, so nothing can be said about that age group (Section D.3).
4. **A thin healthy baseline.** Only 2 NORMAL/NORMAL patients have usable scans in both breasts, so how different two healthy breasts are is barely measured (Appendix D).
5. **The rule selects dense breasts.** BI-RADS C/D (dense) make up 82% of the patients compared against 39% of those left out (Figure D.6). The result applies to dense breasts; for fattier breasts it currently says nothing.
6. **The threshold sits at a peak.** The CANCER vs NORMAL share is 68% at 0.25 (17/25), 83% at 0.35 (15/18), 67% at 0.45 (10/15), 78% at 0.55 (7/9). The chosen 0.35 is at a peak of this curve (Figure D.1), so the 83% is likely the optimistic end, and it was chosen on the same patients it is reported on. The result stays above chance at every threshold. **Mapping would address this:** selection becomes a percentile within each breast after the scan, not a global threshold tuned on this cohort (Chapter 4).

7 Next steps

1. **Label the newest patients.** The newest 45 patients have been put through the same water-peak rule, blind to labels: 52 of 265 scans (20%) are water-rich, and **11 patients** have both breasts usable (Appendix G). That usable rate is lower than the 39% in the main analysis. Once labels are in, those patients can test the CANCER vs NORMAL result. Only 18 of 251 patients in the main analysis ended up in the headline comparison, so this newest group will give a handful of pairs at best — enough to check the direction, not to confirm the 83%.
2. **Strengthen the baseline.** More NORMAL/NORMAL patients with usable scans in both breasts, to measure how different two healthy breasts are.
3. **Keep the threshold fixed** when new labelled patients are added, rather than retuning it, so the peak at 0.35

   (1) is not carried forward.

4. **Trial mapping for Human 2.0.** Run the surface-then-XRD fly-scan protocol (1 Hz detector, speed-set collection, dynamic speed on thicker tissue) and check that a whole-breast map finishes within the 5-minute target (Chapter 4). A simulated half breast at 5 mm steps, 2 million photons (1 s) at each point 2 cm or thicker, is 166 s: 2 min 46 s (Chapter 5).

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13. Ex-vivo X-ray diffraction of breast tissue, Keele University (unpublished data).

A Preparing each scan: peak-shift correction and min-max normalisation

Three example scans, from Nova 250 onwards and one data batch, with a clear fat peak, chosen so their fat peaks start at different q; the red one also has a strong water peak near q = 21. The dots mark each fitted fat peak.

1. **As measured**: the fat peak's position depends on the sample-to-detector distance (SDD), which differs a little between scans, so the peaks don't line up.
2. **Peak-shift corrected**: a Gaussian plus a straight-line baseline is fitted to I(q) between q = 10 and 17.5 nm ^{1}, and the fat peak's position is the Gaussian's centre, so the baseline's slope can't move it. Each scan's SDD is then changed (new SDD = SDD · tan 2 (peak) / tan 2 (14)), the image integrated again and the peak refitted, repeating until the fitted centre is at q = 14 nm ^{1} (`nova_sdd_peak_align.py`). The thick lines in panels 1 and 2 are those fits, the dotted lines their baselines.
3. **Min-max normalised** (Stretch): each corrected curve is rescaled so its trough (the minimum of a cubic over q = 5-10) is 0 and the fit's value at its peak is 1.

The first two panels are in measured units and the third is 0-1, so they share the q axis but not the y axis.

```
EXAMPLE_PEAKS = (12.8, 13.8)   # nm ¹: pick the scans whose measured fat peak is nearest thes
EXAMPLE_BATCH = "none"              # all three from one data batch, so their measured units
MIN_PROMINENCE = 2.5                # fat peak at least this many times the trough, so the pe
WATER_EXAMPLE = (13.1, 0.8)         # a fourth scan with a strong water peak: measured fat pe
Q_START = 1  # nm ¹, where every q axis starts
EXAMPLE_COLOURS = ("tab:blue", "tab:purple", "tab:red")

found = []
for group in ne.stored_scans(ne.STORE_PATH):
    if (group.attrs["why"] or int(group.attrs["nova"]) < 250 or group.attrs["label"] not in n
            or str(group.attrs["batch"]) != EXAMPLE_BATCH or not group.attrs["peak_value"] >=
        continue
    original, adjusted = group["original"], group["adjusted"]
    found.append(dict(name=f"Nova {int(group.attrs['nova'])} {group.attrs['side']}", label=gr
                      centre=float(original.attrs["params"][1]),
```

```
                      q0=original["q"][()], I0=original["I"][()], q1=adjusted["q"][()], I1=ad
                      peak=ne.peak_value(adjusted.attrs["params"]), peak0=ne.peak_value(origi
                      params0=original.attrs["params"], params1=adjusted.attrs["params"],
                      window0=tuple(original.attrs["window"]), window1=tuple(adjusted.attrs["
    stretched = ne.normalised_curve(found[-1]["q1"], found[-1]["I1"], found[-1]["peak"], "str
    found[-1]["water"] = np.nanmean(stretched[(found[-1]["q1"] >= 20) & (found[-1]["q1"] <= 2
def fitted(q_values, params):
    """The stored fit: a Gaussian plus a straight-line baseline whose level applies at ne.FIT
    height, centre, sigma, level, slope = params
    return height * np.exp(-(q_values - centre) ** 2 / (2 * sigma ** 2)) + level + slope * (q

examples = [min(found, key=lambda s: abs(s["centre"] - target)) for target in EXAMPLE_PEAKS]
examples.append(min((s for s in found if s["water"] >= WATER_EXAMPLE[1]), key=lambda s: abs(s

plt.rcParams["font.size"] = 7  # small text so three panels fit side by side on an A4 page
fig, axes = plt.subplots(1, 3, figsize=(8, 3.4), sharex=True)
for ax in axes:
    ax.set_box_aspect(1)  # square panels
for s, colour in zip(examples, EXAMPLE_COLOURS):
    stretched = ne.normalised_curve(s["q1"], s["I1"], s["peak"], "stretch")
    axes[0].plot(s["q0"], s["I0"], color=colour, lw=1.2, label=f"{s['name']} ({s['label']}),
    axes[0].axvline(s["centre"], color=colour, ls="--", lw=1)
    for ax, params, window in ((axes[0], s["params0"], s["window0"]), (axes[1], s["params1"],
        x = np.linspace(*window, 300)
        ax.plot(x, fitted(x, params), color="black", lw=2.2, zorder=4)
        ax.plot(x, fitted(x, params), color=colour, lw=1.2, ls="--", zorder=5)
        ax.plot(x, params[3] + params[4] * (x - ne.FIT_MIDDLE), color=colour, lw=1, ls=":", z
    axes[0].plot(s["centre"], s["peak0"], "o", color=colour, ms=7, mec="black", zorder=6)
    axes[1].plot(14, s["peak"], "o", color=colour, ms=7, mec="black", zorder=6)
    axes[1].plot(s["q1"], s["I1"], color=colour, lw=1.2)
    axes[2].plot(s["q1"], stretched, color=colour, lw=1.2)
axes[1].axvline(14, color="black", ls="--", lw=1, label="q = 14 nm ⁻¹")
fit_key = [Line2D([], [], color="black", lw=2.2, label="fit: Gaussian + straight line, q 10-1
           Line2D([], [], color="grey", lw=1, ls=":", label="its straight-line baseline"),
           Line2D([], [], marker="o", color="grey", mec="black", ls="", label="fitted peak (t
axes[2].axvline(14, color="black", ls="--", lw=1)
axes[2].axhline(0, color="grey", ls=":", lw=1)
axes[2].axhline(1, color="grey", ls=":", lw=1)
axes[2].axvspan(*ne.TROUGH_WINDOW, color="grey", alpha=0.08, label=f"trough window, q {ne.TRO
titles = ("1. As measured:\nfat peaks at different q", "2. Peak-shift corrected:\nevery fat p
ylabels = ("I(q), as measured", "I(q), SDD corrected", "I(q), stretched")
for ax, title, ylabel in zip(axes, titles, ylabels):
```

```
    ax.set(title=title, xlabel="q (nm ⁻¹)", ylabel=ylabel, xlim=(Q_START, 25))
    ax.grid(alpha=0.3)
    handles, _ = ax.get_legend_handles_labels()
    ax.legend(handles=handles + (fit_key if ax is axes[1] else []), fontsize=4.5, loc="upper
for ax in axes[:2]:
    inside = [(s["q0"] if ax is axes[0] else s["q1"]) for s in examples]
    values = np.concatenate([(s["I0"] if ax is axes[0] else s["I1"])[(qq >= Q_START) & (qq <=
    ax.set_ylim(0, np.nanmax(values) * 1.12)  # q up to 23, away from the noisy last points
axes[2].set_ylim(-0.1, 1.45)
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

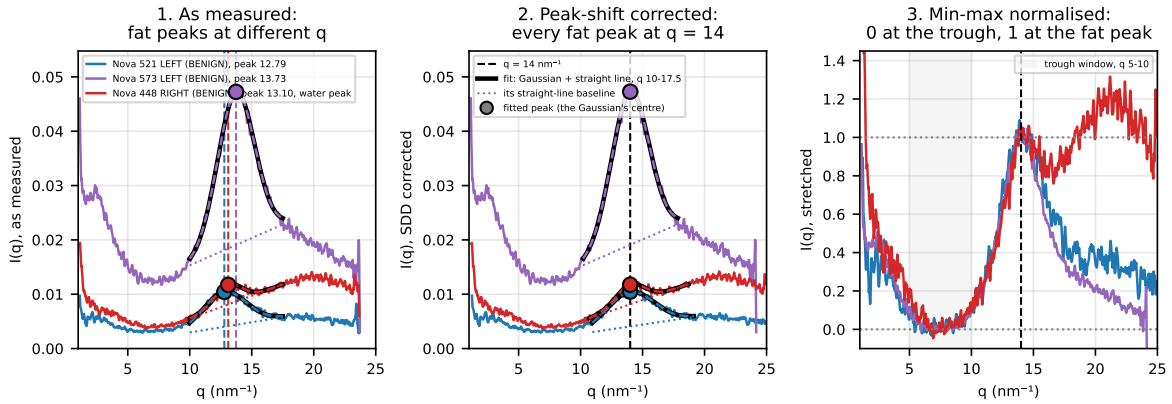


Figure A.1: Peak-shift correction and min-max normalisation of example scans.

i How to read the prepared curves in this document

Because of the min-max normalisation, every fat peak shown in this document has the same intensity: 1, at q = 14 nm ⁻¹. Its height says nothing about the tissue. Focus instead on the **relative size of the broad water peak to the right of it** (around q = 20-22 nm ⁻¹): the higher the water peak relative to the fat peak, the more water and the less fat the beam passed through.

B Measuring Water:Fat: fitting two Gaussians

To compare the two breasts of a patient, each breast gets one number, its **Water:Fat**. All of the breast's scans are peak-shift corrected and min-max normalised (Appendix A), then averaged. A sum of two Gaussians is fitted to that mean curve over q = 7-25 nm ^{1}: one for the fat peak (centre between q = 12 and 16) and one for the broader water peak (centre between q = 17 and 26). The area of each Gaussian within ±3 of its centre is height × × √(2) × erf(3/√2), and the breast's Water:Fat is the water Gaussian's area divided by the fat Gaussian's area. A breast with more water and less fat has a larger Water:Fat, and comparing the two breasts of one patient removes much of what differs from patient to patient.

Figure B.1 shows this for one patient with one NORMAL and one CANCER breast, each breast the mean of all its scans. Under the water-peak rule the mean is of the breast's usable scans only (Appendix C).

```
# FIT_RANGE, the Gaussian fit, water_fat and breast_curves come from cohort_common
# the highest-ranked patient in the top 10% of fig-cohort (right) with one NORMAL and one CA
example = None
for name in [names[k] for k in by_water[:round(len(names) * 0.1)]]:
    breasts_here = breast_curves(ne.row_nova(name))
    if (sorted(label for label, _, _ in breasts_here.values()) == ["CANCER", "NORMAL"]
            and all(fit_two_gaussians(curve) is not None for _, curve, _ in breasts_here.valu
        example, example_breasts = ne.row_nova(name), breasts_here
        break

plt.rcParams["font.size"] = 7  # small text so both panels fit side by side on an A4 page
fig, axes = plt.subplots(1, 2, figsize=(8, 4), sharey=True)
example_ratio = {}
for ax, (side, (label, curve, count)) in zip(axes, example_breasts.items()):
    ax.set_box_aspect(1)  # square panels
    p = fit_two_gaussians(curve)
    fat, water = p[:3], p[3:]
    example_ratio[label] = area_3sigma(*water) / area_3sigma(*fat)
    x = np.linspace(2, 25, 600)
    for (height, centre, sigma), colour, name in ((fat, FAT_COLOUR, "fat"), (water, WATER_CO
        within = (x >= centre - 3 * sigma) & (x <= centre + 3 * sigma)
        ax.fill_between(x[within], gaussian(x[within], height, centre, sigma), color=colour,
```

```
        ax.plot(x, gaussian(x, height, centre, sigma), color=colour, lw=1.5,
                label=f"{name} Gaussian: centre {centre:.1f},   {sigma:.2f}, area {area_3sigm
    ax.plot(q, curve, color="black", lw=1, label=f"mean of {count} scans")
    ax.plot(x, two_gaussians(x, *p), color="red", ls="--", lw=1.5, zorder=5, label="fat + wat
    ax.axvspan(2, FIT_RANGE[0], color="grey", alpha=0.08)  # left out of the fit
    ax.set(title=f"Nova {example} {side} breast: {label}\nWater:Fat = {area_3sigma(*water):.2
           xlabel="q (nm ⁻¹)", xlim=(2, 25))
    ax.set_xticks([5, 10, 15, 20, 25])
    ax.grid(alpha=0.3)
    ax.legend(frameon=False, loc="upper left", fontsize=6)
axes[0].set_ylabel("I(q), stretched")
in_fit = (q >= FIT_RANGE[0]) & (q <= FIT_RANGE[1])
axes[0].set_ylim(0, 1.35 * max(np.nanmax(curve[in_fit]) for _, curve, _ in example_breasts.va
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

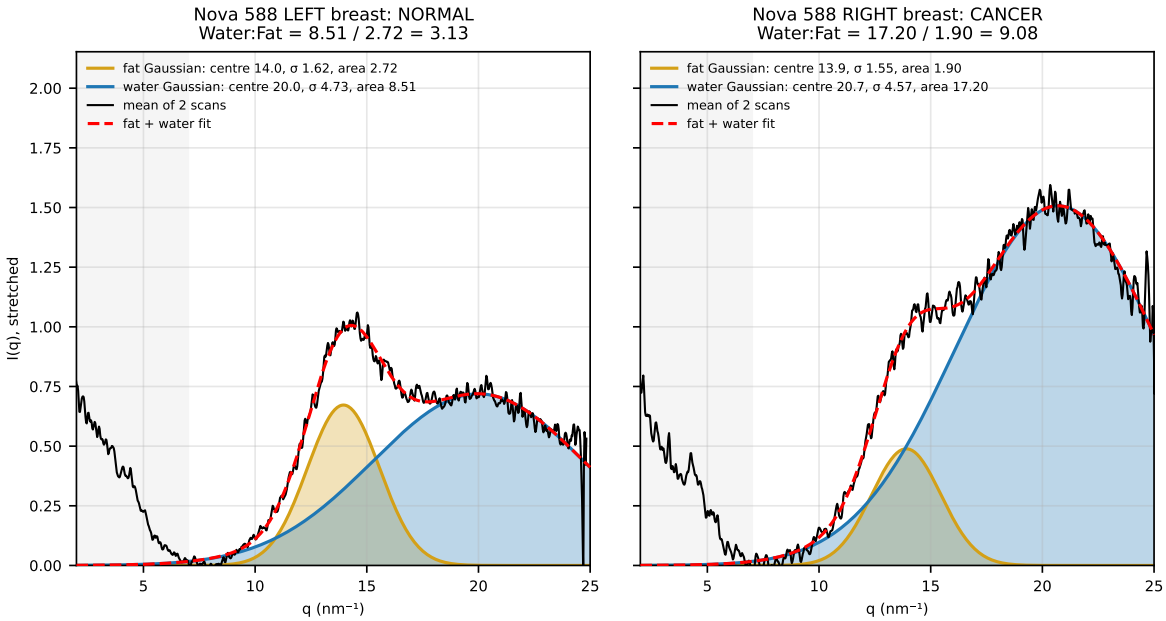


Figure B.1: How Water:Fat is measured, for one CANCER/NORMAL patient. Each panel is one breast: the mean of its scans after preparation (black), the fitted fat Gaussian (yellow) and water Gaussian (blue), each shaded within ±3 of its centre, and their sum (red dashed). Water:Fat is the blue area over the yellow area.

For Nova 588, the CANCER breast's Water:Fat is 9.08 and the NORMAL breast's is 3.13: the

CANCER breast has the higher Water:Fat, by a factor of 2.9.

C The water-peak rule

The filter that decides whether a scan is usable (water-rich). To be a fair test it must meet three conditions:

1. **Blind.** It uses only the scan itself, never the breast's label or the result of the comparison.
2. **Symmetric.** It is applied the same way to both breasts, so neither is favoured.
3. **The same on every patient.** The threshold is not tuned patient by patient. It was, however, chosen by looking at the sensitivity curve on this same set of patients (Figure D.1), so the headline result is not independent of that choice.

Each breast is scanned a few times, usually three, and some of those scans will miss the glandular tissue. So the rule works scan by scan:

- **A scan is usable** if it has a real water peak: its water peak height (the mean of its prepared curve over q = 20-22) is at least **0.35**, that is 35% of the fat peak's height, and its fitted water Gaussian is centred at **q 19**, so the fit has found a water peak and not just the shoulder of the fat peak.
- **A breast is usable** if at least one of its scans is; its Water:Fat comes from the mean of its usable scans only.
- **A patient is compared** only if **both** breasts are usable.

```
found = rule_breasts(WATER_MIN, CENTRE_MIN)
# usable example: the first CANCER/NORMAL patient with both usable and rejected scans in eac
example_nova = next(n for n in analysis_novas
                    if (n, "LEFT") in found and (n, "RIGHT") in found
                    and sorted(found[n, s][0] for s in ("LEFT", "RIGHT")) == ["CANCER", "NOR
                    and all(0 < found[n, s][2] < found[n, s][3] for s in ("LEFT", "RIGHT")))
# not-usable example: a breast with scans but none that pass the rule
rejected_example = next((nova, side, label)
                        for nova, side, label, curve, height, centre in scan_records
                        if label in COHORTS and (nova, side) not in found
                        and sum(1 for n, s, *_ in scan_records if n == nova and s == side) >=

def plot_breast(ax, nova, side, show_fit):
    records = [(label, curve, height, centre, usable(height, centre, WATER_MIN, CENTRE_MIN))
               for n, s, label, curve, height, centre in scan_records if n == nova and s ==
```

```
    label = records[0][0]
    good = []
    for k, (_, curve, height, centre, ok) in enumerate(records):
        tag = f"scan {k + 1}: water {height:.2f}, centre {centre:.1f}"
        if ok:
            good.append(curve)
            ax.plot(q, curve, color=ne.STATUS_COLOURS[label], lw=0.9, label=f"{tag}, usable")
        else:
            ax.plot(q, curve, color="grey", lw=0.8, ls="--", label=f"{tag}, rejected")
    ax.hlines(WATER_MIN, *WATER, color="black", lw=2.5, zorder=6, label=f"threshold: water pe
    if show_fit and good:
        with warnings.catch_warnings():
            warnings.simplefilter("ignore", RuntimeWarning)
            mean_good = np.nanmean(good, axis=0)
        p = fit_two_gaussians(mean_good)
        ratio = water_fat(mean_good)
        x = np.linspace(2, 25, 600)
        for (height, centre, sigma), colour in ((p[:3], FAT_COLOUR), (p[3:], WATER_COLOUR)):
            within = (x >= centre - 3 * sigma) & (x <= centre + 3 * sigma)
            ax.fill_between(x[within], gaussian(x[within], height, centre, sigma), color=colo
        ax.plot(x, two_gaussians(x, *p), color="red", ls="--", lw=1.4, zorder=5, label="fit t
        ax.set_title(f"{side}, {label}: {len(good)} of {len(records)} usable\nWater:Fat = {ra
    else:
        ax.set_title(f"{side}, {label}: 0 of {len(records)} usable\nnot used")
    ax.set(xlabel="q (nm ⁻¹)", xlim=(2, 25))
    ax.set_xticks([5, 10, 15, 20, 25])
    ax.grid(alpha=0.3)
    ax.legend(frameon=False, fontsize=5, loc="upper left")
    ax.set_box_aspect(1)

plt.rcParams["font.size"] = 7
fig, axes = plt.subplots(1, 3, figsize=(9, 3.6), sharey=True)
sides = sorted(("LEFT", "RIGHT"), key=lambda side: found[example_nova, side][0] != "NORMAL")
plot_breast(axes[0], example_nova, sides[0], show_fit=True)
plot_breast(axes[1], example_nova, sides[1], show_fit=True)
plot_breast(axes[2], rejected_example[0], rejected_example[1], show_fit=False)
axes[0].set_ylabel("I(q), prepared")
axes[0].set_ylim(0, None)
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

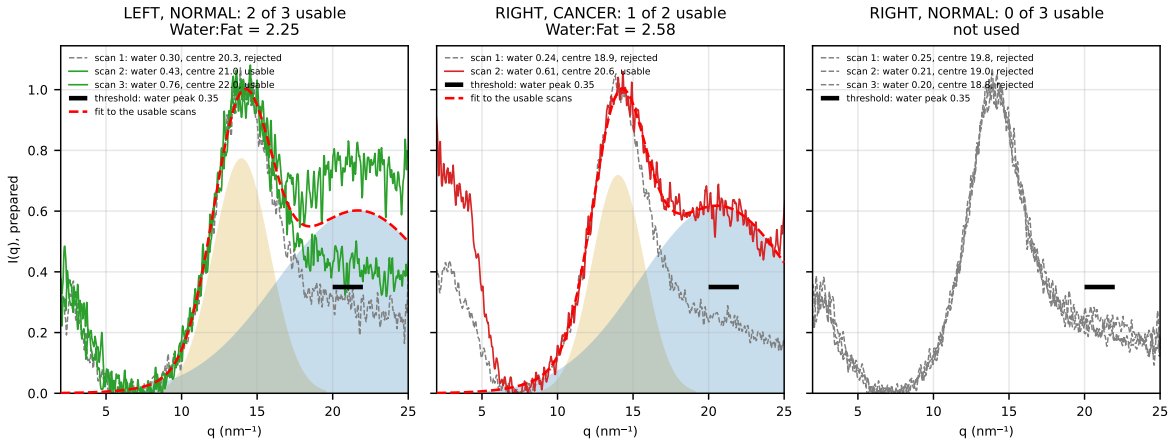


Figure C.1: The rule on three breasts. **Left and middle**: one CANCER/NORMAL patient, each breast with usable scans (coloured) and rejected scans (grey dashed); fat (yellow) and water (blue) Gaussians fitted to the mean of the usable scans, with their sum (red dashed). **Right**: a breast with no usable scan — every scan sits below the threshold (black bar over q 20-22), so the breast is not used.

This fixed threshold is for **single-point** Human 1.0 scans, when only a few beams are placed. It is also the natural **stopgap** on Human 1.0 while mapping is not yet available (Chapter 4): apply the same check during scanning and only keep a position if the short look is water-rich. Chapter 1 (Chapter 1) makes the same "wet = information" point with a top/bottom 15% cut, which is the natural rule once a whole breast is mapped.

Figure C.2 applies the threshold to every breast: rejected (fatty) scans look the same for all three labels; usable scans are wetter for BENIGN and CANCER than for NORMAL, with BENIGN and CANCER almost the same under this cut. The paired comparison that works is CANCER vs NORMAL (Appendix D).

```
rejected_by_label = {lab: [] for lab in COHORTS}
usable_by_label = {lab: [] for lab in COHORTS}
with warnings.catch_warnings():
    warnings.simplefilter("ignore", RuntimeWarning)
    for (nova, side, label), curves in breast_rej.items():
        rejected_by_label[label].append(np.nanmean(curves, axis=0))
    for (nova, side, label), curves in breast_use.items():
        usable_by_label[label].append(np.nanmean(curves, axis=0))

together = np.vstack([np.vstack(c) for c in list(rejected_by_label.values()) + list(usable_by
covered = np.flatnonzero(np.isfinite(together).all(axis=0))
columns = slice(covered[0], covered[-1] + 1)
```

```
mean_water = {}

plt.rcParams["font.size"] = 7
fig, (rej_ax, use_ax) = plt.subplots(1, 2, figsize=(8, 3.8), sharey=True)
for ax, curves_by_label, title in ((rej_ax, rejected_by_label, "Rejected scans"), (use_ax, us
    ax.set_box_aspect(1)
    for label in COHORTS:
        stack = np.vstack(curves_by_label[label])
        with warnings.catch_warnings():
            warnings.simplefilter("ignore", RuntimeWarning)
            mean = np.nanmean(stack, axis=0)
        mean_water[title, label] = float(np.nanmean(mean[(q >= WATER[0]) & (q <= WATER[1])]))
        colour = ne.STATUS_COLOURS[label]
        ax.plot(q[columns], mean[columns], color=colour, lw=1.6,
                label=f"{label}: {len(curves_by_label[label])} breasts, water peak {mean_wate
    ax.axvline(14, color="black", ls="--", lw=0.8)
    ax.hlines(WATER_MIN, *WATER, color="black", lw=2.5, zorder=6)
    ax.set(title=title, xlabel="q (nm ⁻¹)", xlim=(2, q[columns][-1]))
    ax.set_xticks([5, 10, 15, 20])
    ax.grid(alpha=0.3)
    ax.legend(frameon=False, loc="upper right", fontsize=6)
rej_ax.set_ylabel("I(q), prepared (fat peak = 1)")
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

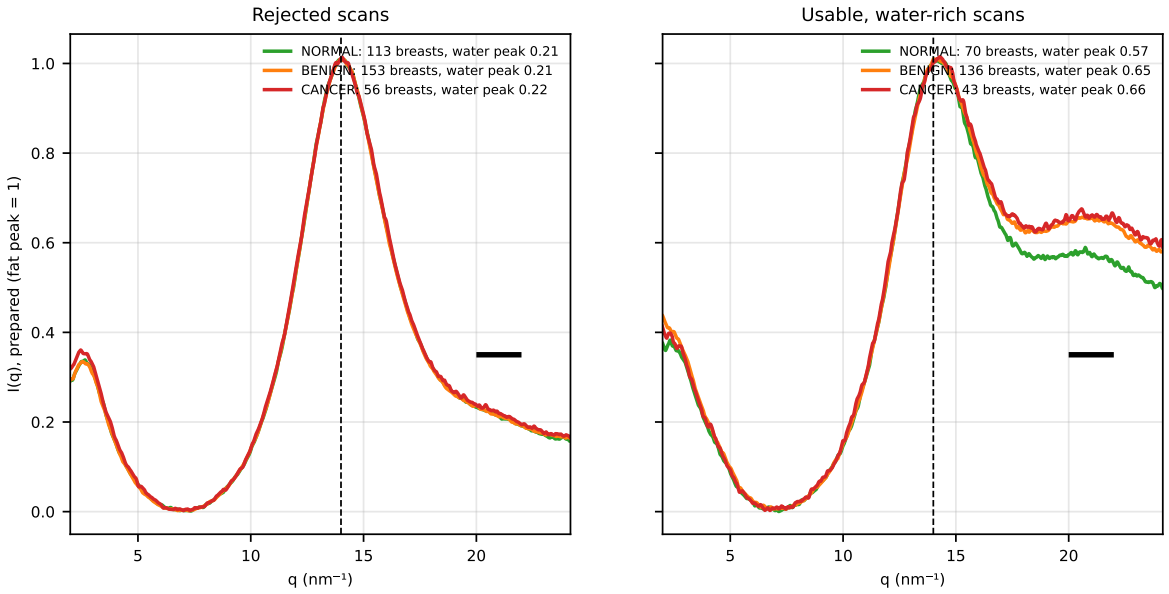


Figure C.2: The mean prepared curve of each cohort, each breast the mean of its rejected scans (**left**) or its usable, water-rich scans (**right**), with the water-peak threshold marked. A breast with both kinds of scan appears in both panels. Rejected scans overlap for all three labels; usable scans are wetter for BENIGN and CANCER than for NORMAL, with BENIGN and CANCER almost the same.

C.1 Before and after

The same paired comparison, one number per patient, using every scan, then only the usable scans, then only the rejected ones:

Scans used	CANCER breast higher	95% CI	p (binomial, against 50%)
Every scan (before)	25/38 (66%)	49%–80%	0.073
Usable scans (after)	**15/18 (83%)**	**59%–96%**	**0.0075**
Rejected scans	11/23 (48%)	27%–69%	1

Using every scan, the CANCER breast is already the wetter one in most patients, but the fatty scans dilute the signal. Taking them out raises the share to 15/18 (83%); the fatty scans on their own are at chance. The rule accepts 39% of all scans. Every patient with both breasts usable under this rule is plotted in Appendix F. Further checks on the threshold and who is left out are in Appendix D.

C.2 Who the rule compares

As in Chapter 1, denser tissue is where the signal is. Under this rule the patients compared are mostly BI-RADS C or D: **82%** of 85, against 39% of the 102 left out (Figure D.6).

D The water-peak filter in detail

The supporting analysis for the water-peak rule of Appendix C: how the result depends on the threshold, how different the two breasts are, what the scans look like by age, and which patients the rule leaves out.

D.1 Every scan against usable scans, and the threshold

```
SMOOTH = np.round(np.arange(0, 0.601, 0.05), 2)  # thresholds for the sensitivity curve; abo
sensitivity = {t: rule_results(t, CENTRE_MIN)["CANCER vs NORMAL"] for t in SMOOTH}

def plot_sensitivity(ax, by_threshold, every_scan):
    """The CANCER vs NORMAL share against the threshold, with the number of patients written
    ts = np.array(list(by_threshold))
    shares = [100 * h / n if n else np.nan for h, n, _ in by_threshold.values()]
    ax.plot(ts, shares, color=ne.STATUS_COLOURS["CANCER"], lw=1.8, marker="o", ms=3, label="
    for t, (h, n, _) in by_threshold.items():
        if n and round(t * 10) == t * 10:
            ax.text(t, 100 * h / n + 3, f"{h}/{n}", ha="center", fontsize=6)
    h, n, _ = every_scan
    ax.axhline(100 * h / n, color="grey", lw=1, label=f"every scan, {h}/{n}")
    ax.axhline(50, color="black", ls=":", lw=1, label="chance")
    ax.axvline(WATER_MIN, color="black", ls="--", lw=1, label=f"chosen threshold, {WATER_MIN
    ax.set(xlabel="minimum water peak of a usable scan (fraction of the fat peak)", ylabel="
           ylim=(0, 110), xlim=(-0.02, SMOOTH[-1] + 0.02))
    ax.grid(alpha=0.3)
    ax.legend(frameon=False, loc="lower left", fontsize=6)
plt.rcParams["font.size"] = 7  # small text so both panels fit side by side on an A4 page
fig, (bar_ax, sens_ax) = plt.subplots(1, 2, figsize=(8, 4))
for ax in (bar_ax, sens_ax):
    ax.set_box_aspect(1)  # square panels
width = 0.38
for k, comparison in enumerate(COMPARISONS):
    for offset, results, colour, name in ((-width / 2, all_scans_results, "lightgrey", "every
```

```
                                            (width / 2, rule, TOP_COLOUR, "usable scans only, h
        higher, total, p_value = results[comparison]
        if not total:
            continue
        bar_ax.bar(k + offset, 100 * higher / total, width=width, color=colour, label=name if
        bar_ax.text(k + offset, 100 * higher / total + 2, f"{higher}/{total}\np={p_value:.2g}
bar_ax.axhline(50, color="black", ls=":", lw=1)
bar_ax.text(len(COMPARISONS) - 0.5, 51, "chance ", ha="right", va="bottom", fontsize=6)
bar_ax.set(xticks=range(len(COMPARISONS)), ylim=(0, 115), yticks=range(0, 101, 20),
           ylabel="patients whose first-named breast\nhas the higher Water:Fat (%)")
bar_ax.set_xticklabels([c.replace(" vs ", "\nvs ") for c in COMPARISONS], fontsize=6)
bar_ax.grid(alpha=0.3, axis="y")
bar_ax.legend(frameon=False, loc="upper center", bbox_to_anchor=(0.5, -0.14), ncols=1, fontsi

plot_sensitivity(sens_ax, sensitivity, all_scans_results["CANCER vs NORMAL"])
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

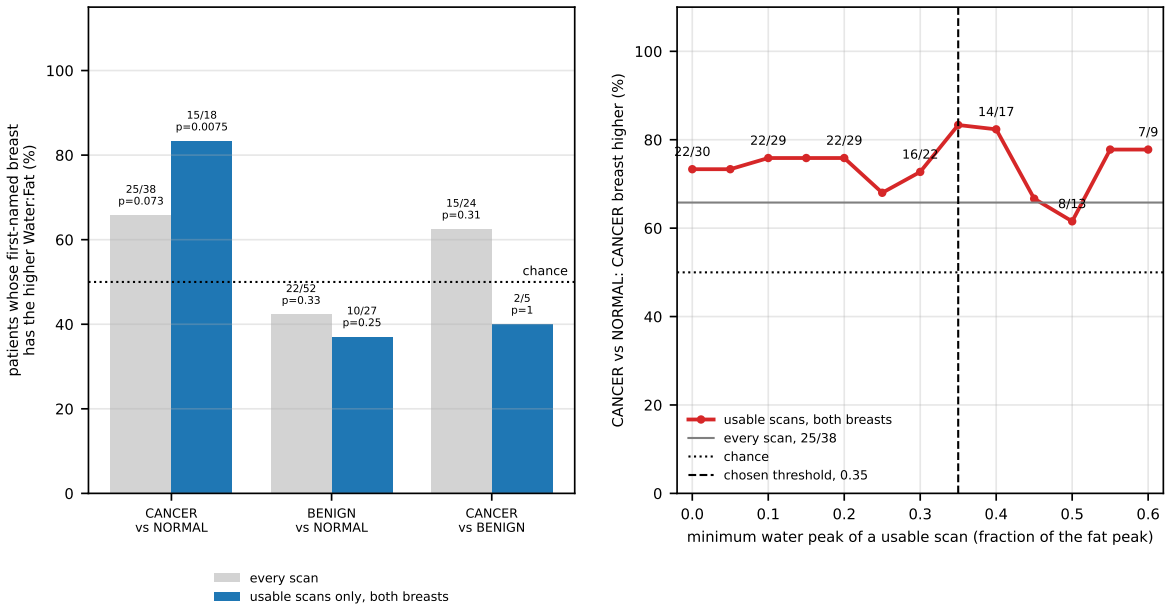


Figure D.1: **Left**: the share of patients whose first-named breast has the higher Water:Fat, using every scan (grey) and using only the scans the rule accepts, with both breasts usable (blue), with the number of patients and the two-sided binomial p-value against 50%. **Right**: the CANCER vs NORMAL result for water peak thresholds from 0 to 0.6 (the water centre rule unchanged), with the patients compared written at every 0.1; the grey line is every scan and the dashed line the chosen threshold.

How much depends on the threshold? The right of Figure D.1 repeats the CANCER vs NORMAL comparison for other thresholds. The result moves around from one threshold to the next, partly because the number of patients falls as the threshold rises; a single threshold should not be read too precisely.

D.2 How different the two breasts are

```
def signed_differences(min_height, min_centre, rejected=False):
    """{comparison: [first-named − second Water:Fat per patient]}, and {"NORMAL vs NORMAL":
    found = rule_breasts(min_height, min_centre, rejected=rejected)
    out = defaultdict(list)
    for nova in sorted({n for n, _ in found if (n, "LEFT") in found and (n, "RIGHT") in found
        (label_a, ratio_a, *_), (label_b, ratio_b, *_) = found[nova, "LEFT"], found[nova, "RI
        if not (np.isfinite(ratio_a) and np.isfinite(ratio_b)):
```

```
            continue
        for comparison, (base, others) in COMPARISONS.items():
            if label_a == base and label_b in others:
                out[comparison].append(ratio_b - ratio_a)
            elif label_b == base and label_a in others:
                out[comparison].append(ratio_a - ratio_b)
        if label_a == label_b == "NORMAL":
            out["NORMAL vs NORMAL"].append(ratio_a - ratio_b)
    return out

signed_rule = signed_differences(WATER_MIN, CENTRE_MIN)
signed_rejected = signed_differences(WATER_MIN, CENTRE_MIN, rejected=True)

rng = np.random.default_rng(1)
plt.rcParams["font.size"] = 7  # small text so both panels fit side by side on an A4 page
fig, (signed_ax, size_ax) = plt.subplots(1, 2, figsize=(8, 4.2))
for ax in (signed_ax, size_ax):
    ax.set_box_aspect(1)  # square panels
names_signed = list(COMPARISONS) + ["NORMAL vs NORMAL"]  # NN is LEFT - RIGHT
names_size = list(COMPARISONS) + ["NORMAL vs NORMAL"]
for ax, names_here, transform in ((signed_ax, names_signed, lambda v: v), (size_ax, names_si
    groups = []
    for k, name in enumerate(names_here):
        groups.append((k - 0.2, transform(signed_rule.get(name, [])), USABLE_COLOUR))
        groups.append((k + 0.2, transform(signed_rejected.get(name, [])), REJECTED_COLOUR))
    box_with_points(ax, groups, rng)
    ax.set_xticks(range(len(names_here)))
    ax.set_xticklabels([f"{n.replace(' vs ', chr(10) + 'vs ')}\n(n = {len(signed_rule.get(n,
    ax.grid(alpha=0.3, axis="y")
signed_ax.axhline(0, color="black", lw=1)
signed_ax.set(ylabel="first-named - second, Water:Fat", title="Which breast has more water")
size_ax.set(ylabel="|A - B|, Water:Fat", title="How different the two breasts are", ylim=(0,
from matplotlib.patches import Patch
signed_ax.legend(handles=[Patch(color=USABLE_COLOUR, alpha=0.5, label="usable scans"), Patch
                 frameon=False, loc="upper center", bbox_to_anchor=(1.1, -0.2), ncols=2, fon
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

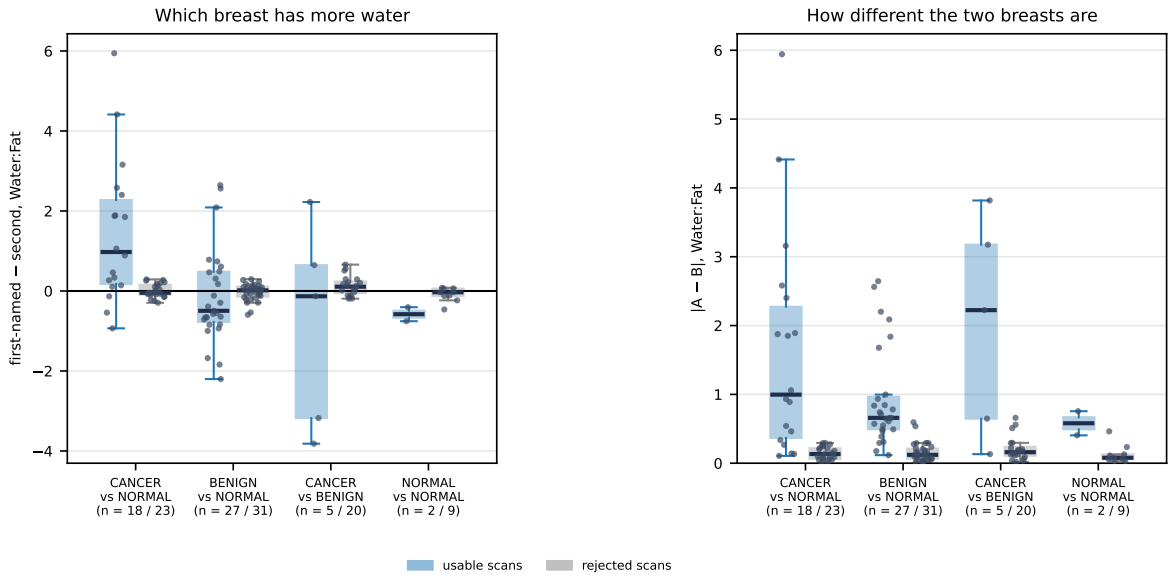


Figure D.2: Each patient's two breasts compared, using only the usable scans (blue) or only the rejected scans (grey); one dot per patient. **Left**: the signed difference, first-named − second (CANCER − NORMAL, etc.; NORMAL vs NORMAL is LEFT − RIGHT); above 0, the first-named breast has more water. **Right**: |A − B|, which does not need to know which breast is which. Boxes span the middle half of the patients, the thick line is the median, and the whiskers reach the furthest patient within 1.5 box-lengths.

The rule accepts 39% of all scans. The results using **only the usable scans**:

Usable scans	First-named breast higher	Median difference between the breasts, |A − B|
CANCER vs NORMAL	**15/18 (83%)**	1.00
BENIGN vs NORMAL	**10/27 (37%)**	0.66
CANCER vs BENIGN	**2/5 (40%)**	2.22
NORMAL vs NORMAL	–	**0.58** (2 patients)

and the same comparisons using **only the scans the rule rejects**, the other 61% (each breast is the mean of its rejected scans, and a patient is compared only if both breasts have one):

Rejected scans	First-named breast higher	Median difference between the breasts, |A − B|
CANCER vs NORMAL	**11/23 (48%)**	0.13

Rejected scans	First-named breast higher	Median difference between the breasts, |A − B|
BENIGN vs NORMAL	**16/31 (52%)**	0.12
CANCER vs BENIGN	**14/20 (70%)**	0.16
NORMAL vs NORMAL	–	**0.08** (9 patients)

The **difference between the breasts** is |A − B| of the two breasts' Water:Fat, the median over the patients in each row. Because every curve is already normalised so its fat peak is 1, the absolute gap has a direct meaning and needs no further scaling. It does not need to know which breast is which, so it is what a blind measurement would see. **NORMAL vs NORMAL** is the baseline: how different two healthy breasts of the same patient are. If a cancer is to be spotted blind, a CANCER/NORMAL pair has to differ by clearly more than that.

From the usable scans, the CANCER breast is the higher in **15/18 (83%)** of CANCER/NORMAL patients, and the two breasts differ by a median of **1.00**. From the rejected, fatty scans it is the higher in only **11/23 (48%)**, about what chance would give, and the breasts differ by **0.13**, close to the **0.08** of two NORMAL breasts. Fatty scans cannot tell a CANCER breast from a NORMAL one; the information is in the water-rich scans. The NORMAL vs NORMAL baseline is thin, though: 2 patients have two NORMAL breasts with usable scans and 9 with rejected scans.

D.3 Usable and rejected scans by age

Breasts are averaged within each age band, separately for NORMAL, BENIGN and CANCER, twice: once using only the scans that pass the test (usable), and once using only the scans it rejects. In each, a breast is the mean of the scans used. This shows what the kept and the discarded scans look like at different ages.

```
plot_bands(summaries["usable"])
```

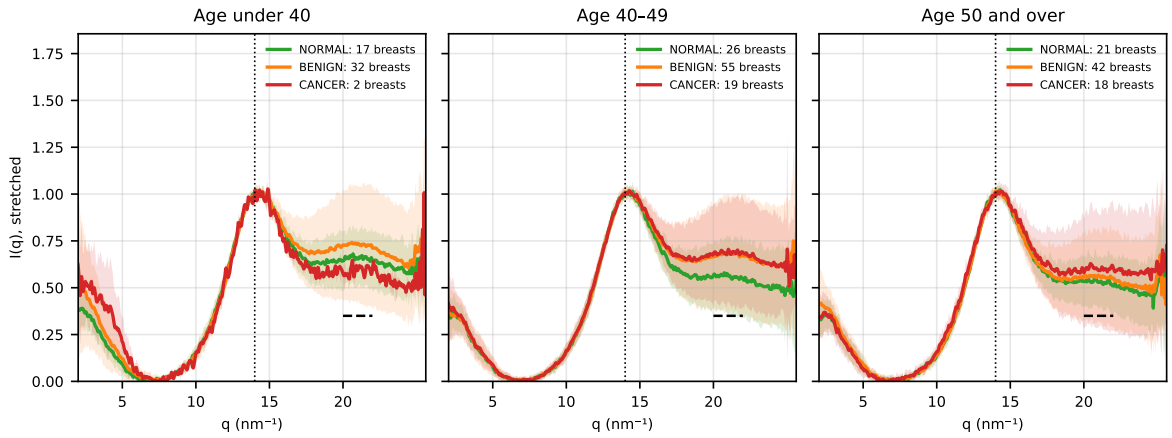


Figure D.3: **Usable scans.** Mean prepared curve of NORMAL, BENIGN and CANCER breasts, one panel per age band (each breast is the mean of its usable scans), with ±1 SD across breasts shaded. The dashed line is the usable-scan water threshold over q 20–22.

```
plot_bands(summaries["rejected"])
```

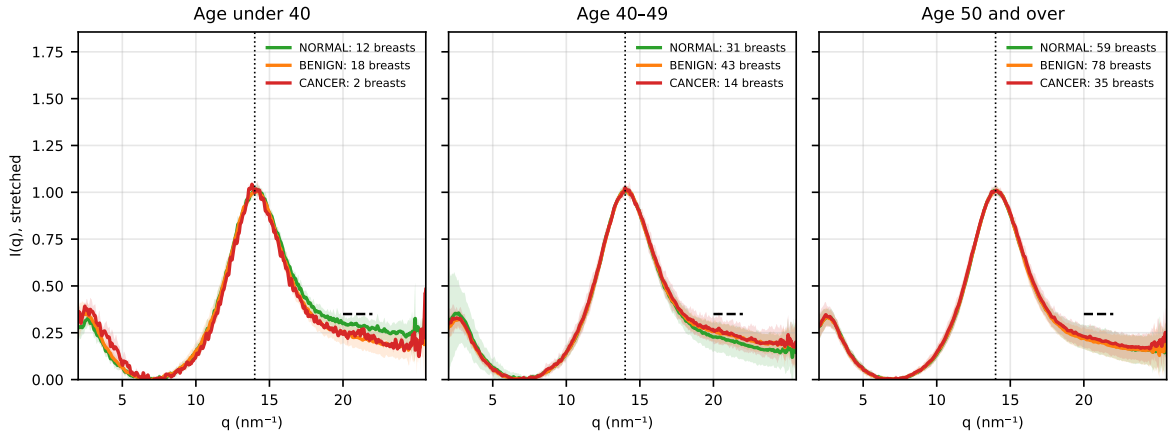


Figure D.4: **Rejected scans.** As Figure D.3, but each breast is the mean of the scans the test rejects.

The table below sums this up: the mean water peak height (q 20–22) of each band's mean curve, with the number of breasts in brackets, each cell coloured on the heatmaps' scale (yellow high, dark blue low):

Cohort

Scans used

under 40

40–49

50 and over

NORMAL

usable

0.66 (17)

0.56 (26)

0.53 (21)

NORMAL

rejected

0.29 (12)

0.21 (31)

0.21 (59)

BENIGN

usable

0.73 (32)

0.68 (55)

0.56 (42)

BENIGN

rejected

0.22 (18)

0.24 (43)

0.20 (78)

CANCER

usable

0.59 (2)

0.69 (19)

0.60 (18)

CANCER

rejected

0.24 (2)

0.25 (14)

0.21 (35)

Breasts without an age are left out, and both figures share one y axis. The rejected scans look the same at every age and in every cohort, with a water peak near 0.2: once the water-bearing scans are taken out, what is left is fat, whatever the patient's age or label. The usable scans fall only a little with age, and CANCER stays about level while NORMAL and BENIGN drop, so in the older bands CANCER is the wettest. Only 2 CANCER breasts are under 40, so those curves say little.

D.4 Water peak against age

Figure D.5 checks whether water peak height among **all usable** breasts is linked to age — NORMAL, BENIGN and CANCER on one axes, each in its cohort colour. One point per breast; a least-squares line and Spearman per label.

```
import h5py
from scipy.stats import spearmanr

# age is stored per scan in the HDF5; one value per Nova
_nova_age = {}
with h5py.File(ne.STORE_PATH, "r") as _store:
    for _group in _store["scans"].values():
        _nova = int(_group.attrs["nova"])
        _age = float(_group.attrs.get("age", np.nan))
        if np.isfinite(_age):
            _nova_age[_nova] = _age

def _age_water_points(label):
    ages, waters = [], []
    for (nova, side, lab), curves in breast_use.items():
        if lab != label or nova not in _nova_age:
            continue
        with warnings.catch_warnings():
            warnings.simplefilter("ignore", RuntimeWarning)
            mean_curve = np.nanmean(curves, axis=0)
            water_h = float(np.nanmean(mean_curve[(q >= WATER[0]) & (q <= WATER[1])]))
```

```
        if not np.isfinite(water_h):
            continue
        ages.append(_nova_age[nova])
        waters.append(water_h)
    return np.asarray(ages, float), np.asarray(waters, float)

_age_water_by_label = {lab: _age_water_points(lab) for lab in COHORTS}
_age_water_stats = {lab: spearmanr(ages, waters) for lab, (ages, waters) in _age_water_by_lab
                    if len(ages) >= 2}

plt.rcParams["font.size"] = 7
fig, ax = plt.subplots(figsize=(4.8, 3.8))
ax.set_box_aspect(1)
for label in COHORTS:
    ages, waters = _age_water_by_label[label]
    if len(ages) < 2:
        continue
    rho, p_val = _age_water_stats[label]
    colour = ne.STATUS_COLOURS[label]
    ax.scatter(ages, waters, s=18, color=colour, alpha=0.75, lw=0, zorder=3,
               label=f"{label} (n = {len(ages)},   = {rho:.2f})")
    fit = np.polyfit(ages, waters, 1)
    line_x = np.linspace(ages.min(), ages.max(), 50)
    ax.plot(line_x, np.polyval(fit, line_x), color=colour, lw=1.4, zorder=2)
ax.axhline(WATER_MIN, color="black", lw=1, alpha=0.45, ls="--")
ax.set(title="Usable breasts: water peak vs age",
       xlabel="Age (years)", ylabel="Water peak height (q 20-22)")
ax.grid(alpha=0.3)
ax.legend(frameon=False, loc="upper right", fontsize=6)
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

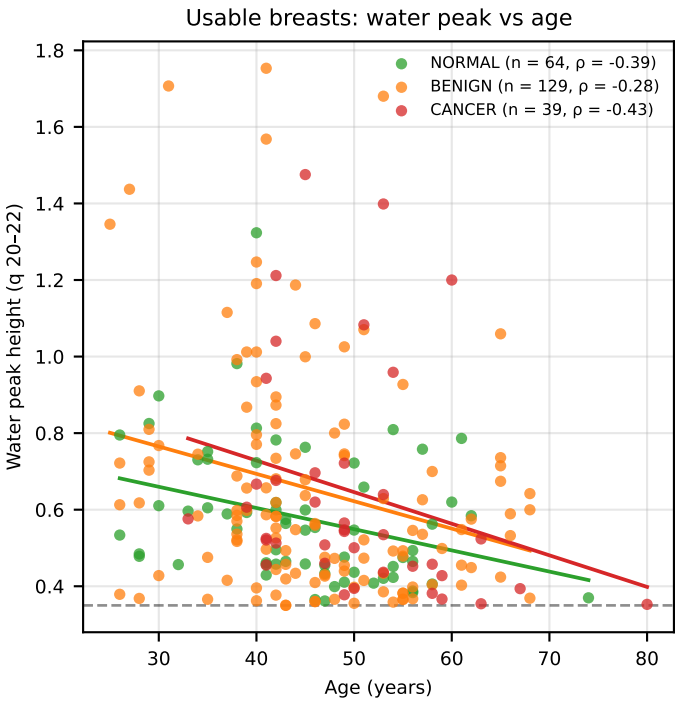


Figure D.5: All usable breasts: water peak height (mean over q 20–22 of the breast's usable-scan mean) against patient age. Points coloured by label; solid line is ordinary least squares within that label; dashed line is the usable-scan water threshold.

In every usable cohort the link runs the same way — **older → lower water peak (fattier)** — Spearman = -0.39 NORMAL (p = 0.0017), -0.28 BENIGN (p = 0.0015), -0.43 CANCER (p = 0.0066).

D.5 Who the rule leaves out

Who are the patients the rule cannot compare? Of the 187 patients with both breasts scanned (NORMAL, BENIGN or CANCER), the rule compares **85** and leaves out **102**: 37 with one breast without a usable scan and 65 with neither. The pathology and radiology sheets in `Images/` were searched for each patient, independently of the scattering data.

```
plt.rcParams["font.size"] = 7
fig, (density_ax, age_ax) = plt.subplots(1, 2, figsize=(8, 4.6))
for ax in (density_ax, age_ax):
    ax.set_box_aspect(1)
width = 0.38
```

```
for offset, group, colour in ((-width / 2, "compared", USABLE_COLOUR), (width / 2, "left out"
    shares = [100 * density_share[group][c] for c in DENSITY_ORDER]
    density_ax.bar(np.arange(len(DENSITY_ORDER)) + offset, shares, width=width, color=colour
                   label=f"{group} ({len(rule_groups[group])} patients)")
    for k, s in enumerate(shares):
        density_ax.text(k + offset, s + 1, f"{s:.0f}", ha="center", va="bottom", fontsize=5.5
density_ax.set_xticks(range(len(DENSITY_ORDER)))
density_ax.set_xticklabels(DENSITY_ORDER, fontsize=5.5, rotation=35, ha="right", rotation_mod
density_ax.set(ylabel="patients (%)", title="Breast density in the radiology reports", ylim=
density_ax.grid(alpha=0.3, axis="y")
density_ax.legend(frameon=False, fontsize=6, loc="upper left")
box_with_points(age_ax, [(0, group_ages["compared"], USABLE_COLOUR), (1, group_ages["left out
age_ax.set_xticks([0, 1])
age_ax.set_xticklabels([f"compared\n(n = {len(group_ages['compared'])})", f"left out\n(n = {l
age_ax.set(ylabel="age (years)", title=f"Age (Mann-Whitney p = {age_p:.1g})")
age_ax.grid(alpha=0.3, axis="y")
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

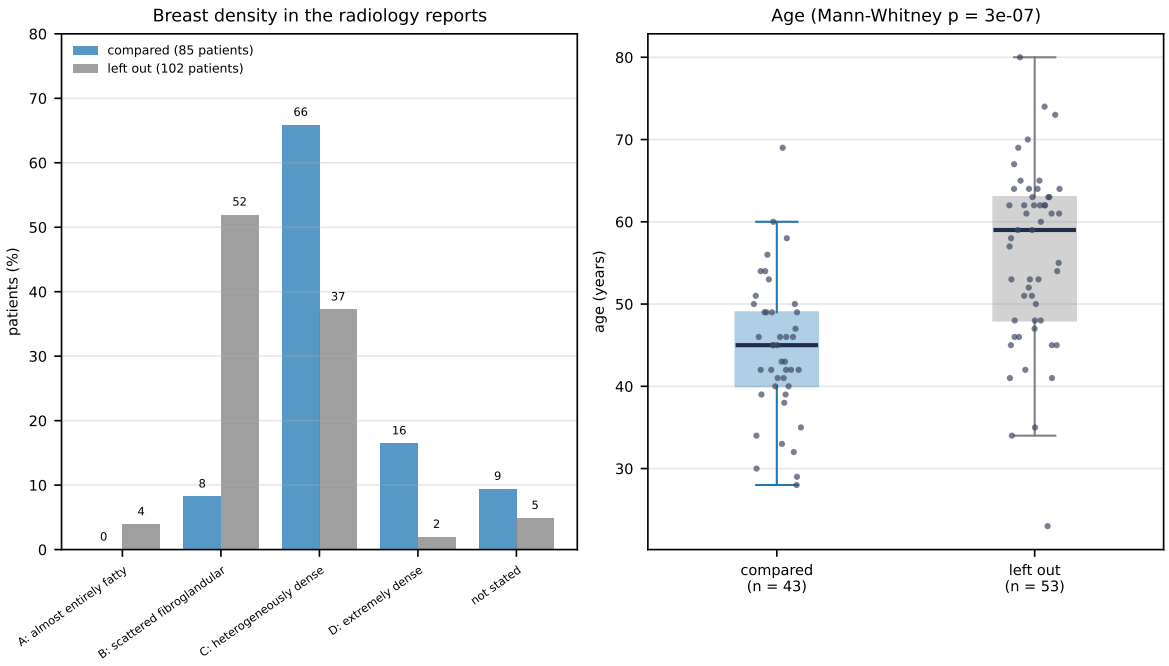


Figure D.6: Patients the rule compares (blue) and leaves out (grey). **Left**: BI-RADS breast density, the category the radiology reports mention most often for each patient. **Right**: age, where the sheets give it; one dot per patient.

The pattern is clear (Figure D.6): **the rule leaves out the fatty breasts.** Radiologists describe a mostly fatty breast as having "scattered areas of fibroglandular density" (BI-RADS B); that is the main description for **52%** of the patients left out but only **8%** of those compared. The patients compared are mostly heterogeneously or extremely dense (BI-RADS C/D: 82%, against 39% of those left out). They are also younger, median **45** against **59** years, as expected since breasts become fattier with age.

Mentioned in the patient's records	Compared	Left out	p (Fisher)
"scattered areas of fibroglandular density" (mostly fatty)	8%	54%	1e-11
"dense" (any)	85%	49%	3e-07
"extremely dense"	18%	3%	0.0008
"palpable"	41%	22%	0.004
"screening (asymptomatic)"	25%	38%	0.06

Mentioned in the patient's records	Compared	Left out	p (Fisher)
"calcification"	76%	89%	0.03
"carcinoma"	29%	40%	0.2
"mastitis"	8%	2%	0.08
"family history"	4%	2%	0.7

Patients who came in with a palpable lump, often younger women with dense breasts, are more often compared; routine screening patients lean towards being left out. Mastitis and family history make no clear difference.

i What this means

1. **The rule is, in effect, a breast-density filter.** Its scan-by-scan water-peak check, which never sees the clinical records, keeps mostly BI-RADS C/D breasts and drops mostly B. This confirms from independent records that rejected scans come from fatty tissue, not from bad measurements.
2. **This is where it would be most useful.** Dense breasts are where mammography is weakest (the reports here repeatedly say density "may obscure small masses"), so a method that works in dense breasts would complement it.
3. **It narrows who the result applies to.** The CANCER vs NORMAL result of the rule holds for younger women with dense breasts; for the older, fattier group it currently says nothing.
4. **Age cannot bias the comparison itself**: both breasts of a patient have the same age, so age decides only *who* is compared, not which breast comes out higher.

D.5.1 Is it just the odds of hitting glandular tissue?

A dense breast holds more glandular tissue, so a scan placed anywhere in it is more likely to pass through some. The table shows, for each density, how often a scan is usable, how often a patient ends up with both breasts usable, and how often a breast's scans disagree (some usable, some rejected):

Density	Patients	Scans usable	Patients with both breasts usable	Breasts whose scans disagree
A: almost entirely fatty	4	0%	0%	0%
B: scattered fibroglandular	60	11%	12%	16%

Density	Patients	Scans usable	Patients with both breasts usable	Breasts whose scans disagree
C: heterogeneously dense	94	49%	60%	40%
D: extremely dense	16	82%	88%	16%
not stated	13	53%	62%	31%

Density sets the odds. A single scan is usable in 11% of cases in mostly fatty (B) breasts and 82% in extremely dense (D) ones. That is why the rule leaves out the fatty breasts.

A breast's scans usually agree. Most breasts give all usable or all rejected scans. If each scan were an independent coin flip, about 71% of C breasts would have a mix; only 40% do. The scans are taken close together and sample the same region, so if a scan misses, moving to a clearly different spot should help more than repeating nearby.

Age matters even at the same density:

Density	Chance a scan is usable, age < 50	age 50
B: scattered fibroglandular	29% (7 patients)	7% (24 patients)
C: heterogeneously dense	58% (28 patients)	29% (19 patients)

Within the same BI-RADS category, younger patients' scans are usable two to four times as often. Either BI-RADS, a coarse four-step scale, hides real differences in density, or glandular tissue itself holds less water with age (it involutes after menopause, with fewer cells and more collagen), so even a scan that does hit it may not show a water peak above the threshold. This data cannot separate the two, and the groups are small.

E Breasts too fatty to measure

The rule of Appendix C compares a patient only if both breasts have a usable scan. A breast with no usable scan is not mismeasured; it is left out because every spot the beam hit was fat. Many of these breasts are genuinely NORMAL, and being too fatty to measure is itself a result: that breast has little water. This section looks at those breasts, and at what happens if the test counts them as the fattier breast instead of leaving the patient out.

E.1 Who has no usable scan

Breasts with no usable scan, by label:

Label	Breasts	No usable scan
NORMAL	138	68 (49%)
BENIGN	224	88 (39%)
CANCER	76	33 (43%)

And patients by how many of their breasts have a usable scan:

Patients	All	Both breasts usable	One	Neither
CANCER vs NORMAL	38	18	5	15 (39%)
BENIGN vs NORMAL	52	27	6	19 (37%)
CANCER vs BENIGN	24	5	9	10 (42%)
NORMAL vs NORMAL	10	2	0	8 (80%)

Of the 10 patients with two NORMAL breasts, 8 have no usable scan in either breast, so the rule never sees them (Figure E.1). These are the most typical healthy patients in the study, and their breasts are fatty all the way through.

```
SIDE_COLOURS = {"LEFT": "tab:blue", "RIGHT": "tab:orange"}
curves = [curve for n in nn_neither for side in ("LEFT", "RIGHT") for curve, _ in breast_sca
covered = np.flatnonzero(np.isfinite(np.vstack(curves)).all(axis=0))
columns = slice(covered[0], covered[-1] + 1)
```

```
y_top = 1.05 * np.nanmax(np.vstack(curves)[:, columns])

plt.rcParams["font.size"] = 7
n_cols = 4
n_rows = int(np.ceil(len(nn_neither) / n_cols))
fig, axes = plt.subplots(n_rows, n_cols, figsize=(8, 2.1 * n_rows), sharex=True, sharey=True
for ax, nova in zip(axes.flat, nn_neither):
    for side, colour in SIDE_COLOURS.items():
        for curve, _ in breast_scans[nova, side]:
            ax.plot(q[columns], curve[columns], color=colour, lw=0.9, alpha=0.8)
    ax.hlines(WATER_MIN, *WATER, color="black", lw=2)
    age = f", age {nova_age[nova]:.0f}" if nova in nova_age else ""
    ax.set_title(f"Nova {nova}{age}")
    ax.set(xlim=(2, q[columns][-1]), ylim=(0, y_top))
    ax.grid(alpha=0.3)
for ax in axes.flat[len(nn_neither):]:
    ax.axis("off")
for ax in axes[-1]:
    ax.set_xlabel("q (nm ⁻¹)")
for ax in axes[:, 0]:
    ax.set_ylabel("I(q), stretched")
axes[0, 0].legend(handles=[Line2D([], [], color=c, label=s.lower()) for s, c in SIDE_COLOURS
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

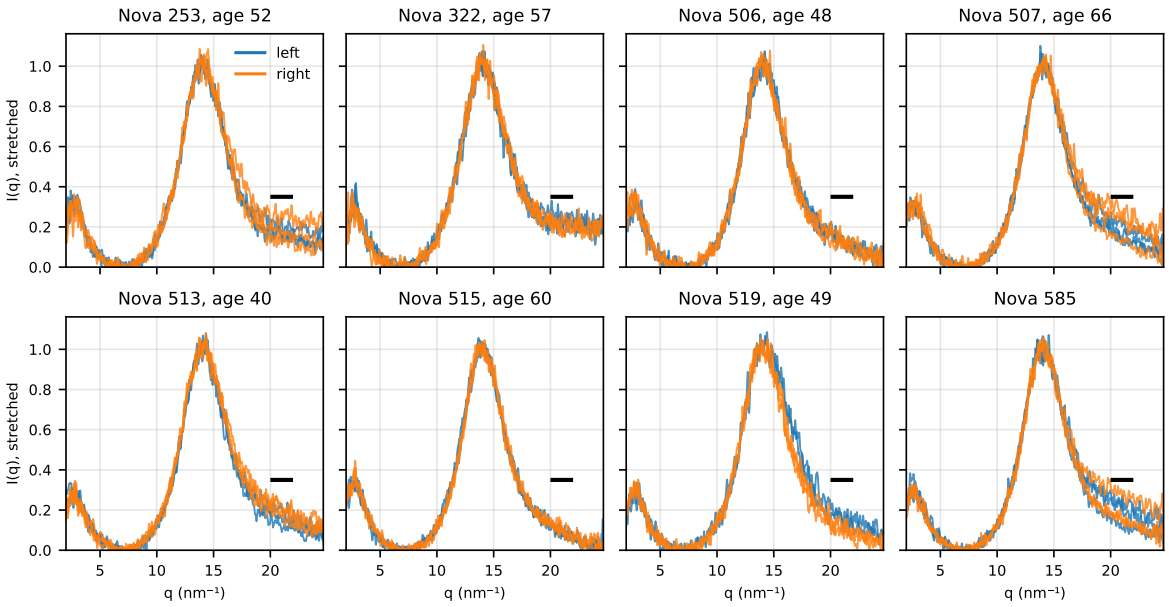


Figure E.1: The NORMAL/NORMAL patients in which neither breast has a usable scan. Each panel is one patient: every prepared scan of the left (blue) and right (orange) breast, with the usable-scan water threshold over q 20–22 (black bar). The water peak never reaches the threshold.

E.2 Counting a fatty breast as a result

If a breast has no usable scan, the test can treat it as the fattier breast of the pair instead of dropping the patient. A patient whose two breasts both have no usable scan is still left out: there is nothing to tell them apart.

Comparison	Both breasts usable	Counting a breast with no usable scan as the fattier
CANCER vs NORMAL	15/18 (83%), p = 0.008	18/23 (78%), p = 0.011
BENIGN vs NORMAL	10/27 (37%), p = 0.248	11/33 (33%), p = 0.080
CANCER vs BENIGN	2/5 (40%), p = 1.000	8/14 (57%), p = 0.791

For CANCER vs NORMAL this adds the patients where only the CANCER breast has a usable scan, which count as right, and those where only the NORMAL breast has one, which count as wrong. The first kind outnumber the second, so more patients are compared with about the same strength of evidence. For BENIGN vs NORMAL it goes the other way: more often it is

the BENIGN breast that has no usable scan, and the result falls further below chance. Most of the patients the rule drops are not lost wins, as in most of them both breasts are fatty.

Having two fatty breasts is much more common in NORMAL/NORMAL patients (80%) than in any other kind of patient (37–42%). So "too fatty to measure in both breasts" leans towards a healthy patient. NORMAL/NORMAL patients are older, though (median age 54 against 49 for the other patients), and fatty breasts are common with age, so this may be age as much as health. With only 10 NORMAL/NORMAL patients it is a hint, not a result.

F Patients under the water-peak rule

Chapter 1 (Chapter 1) is the **top/bottom 15%** cut on samples. This appendix is different: it applies the **fixed water-peak rule** of Appendix C (threshold 0.35) scan by scan, and shows every patient in the analysis set with **both breasts usable** under that rule.

Plotted the same way as the newest patients (Appendix G): one panel per patient, usable scans coloured (left blue, right orange), rejected scans grey and dashed, black bar the water-peak threshold. No outcome tinting — every patient in the usable groups already has both breasts usable. Grouped by the label pair used in the paired comparisons (109 patients), plus the NORMAL/NORMAL patients that missed the threshold entirely.

F.1 CANCER vs NORMAL

18 patients: the paired comparison under the rule of Appendix C.

```
plot_group(group_novas["CANCER vs NORMAL"], "CANCER vs NORMAL")
```

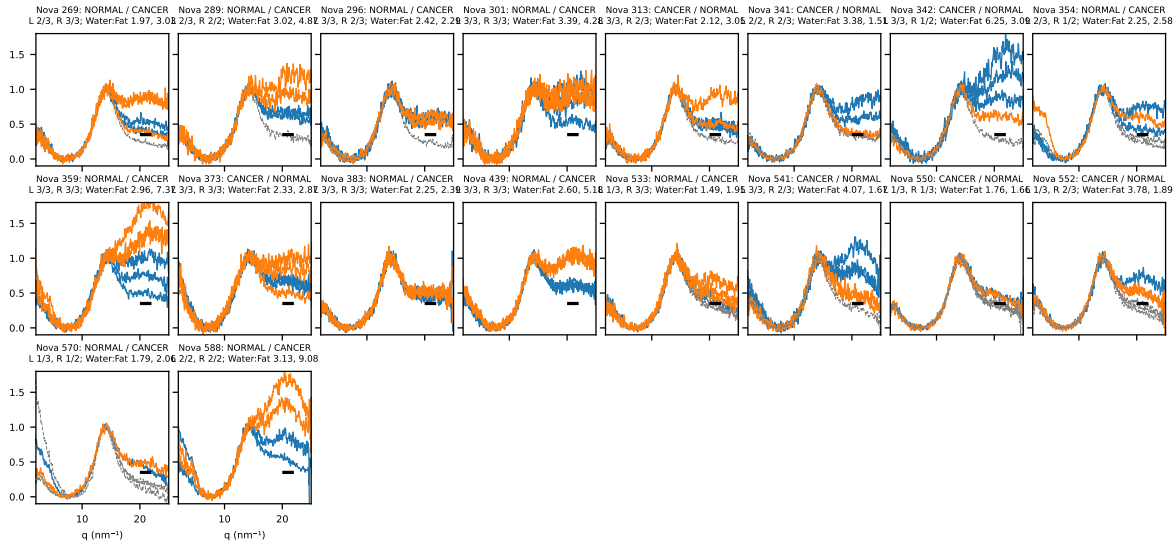


Figure F.1: CANCER vs NORMAL patients with both breasts usable. Usable scans coloured (left blue, right orange), rejected scans grey dashed; black bar is the water-peak threshold.

Patient	Labels (L / R)	Usable scans, L	Usable scans, R	Water:Fat, L	Water:Fat, R
Nova 269	NORMAL / CANCER	2 of 3	3 of 3	1.97	3.03
Nova 289	NORMAL / CANCER	2 of 3	2 of 2	3.02	4.87
Nova 296	NORMAL / CANCER	3 of 3	2 of 3	2.42	2.29
Nova 301	NORMAL / CANCER	3 of 3	3 of 3	3.39	4.28
Nova 313	CANCER / NORMAL	3 of 3	2 of 3	2.12	3.05
Nova 341	CANCER / NORMAL	2 of 2	2 of 3	3.38	1.51
Nova 342	CANCER / NORMAL	3 of 3	1 of 2	6.25	3.09
Nova 354	NORMAL / CANCER	2 of 3	1 of 2	2.25	2.58
Nova 359	NORMAL / CANCER	3 of 3	3 of 3	2.96	7.37

Patient	Labels (L / R)	Usable scans, L	Usable scans, R	Water:Fat, L	Water:Fat, R
Nova 373	CANCER / NORMAL	3 of 3	3 of 3	2.33	2.87
Nova 383	NORMAL / CANCER	3 of 3	3 of 3	2.25	2.39
Nova 439	NORMAL / CANCER	3 of 3	3 of 3	2.60	5.18
Nova 533	NORMAL / CANCER	1 of 3	3 of 3	1.49	1.95
Nova 541	CANCER / NORMAL	3 of 3	2 of 3	4.07	1.67
Nova 550	CANCER / NORMAL	1 of 3	1 of 3	1.76	1.66
Nova 552	CANCER / NORMAL	1 of 3	2 of 3	3.78	1.89
Nova 570	NORMAL / CANCER	1 of 3	1 of 2	1.79	2.06
Nova 588	NORMAL / CANCER	2 of 2	2 of 2	3.13	9.08

F.2 BENIGN vs NORMAL

27 patients.

```
plot_group(group_novas["BENIGN vs NORMAL"], "BENIGN vs NORMAL")
```

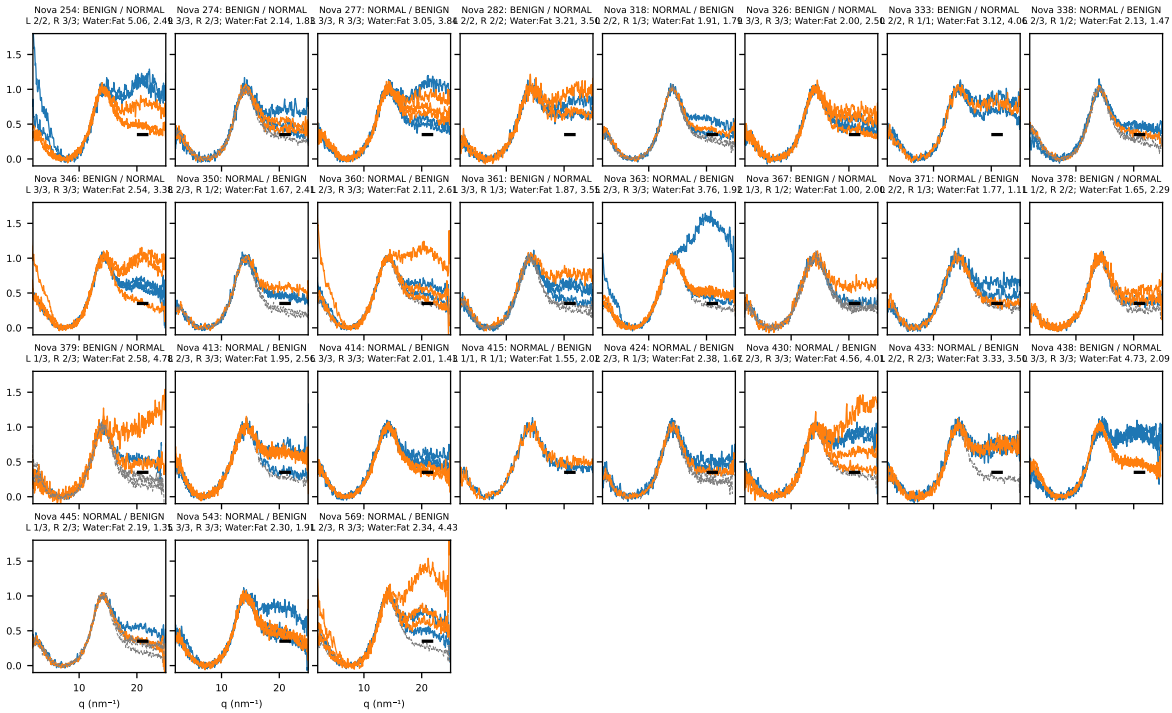


Figure F.2: BENIGN vs NORMAL patients with both breasts usable. Same layout as Figure F.1.

Patient	Labels (L / R)	Usable scans, L	Usable scans, R	Water:Fat, L	Water:Fat, R
Nova 254	BENIGN / NORMAL	2 of 2	3 of 3	5.06	2.49
Nova 274	BENIGN / NORMAL	3 of 3	2 of 3	2.14	1.83
Nova 277	NORMAL / BENIGN	3 of 3	3 of 3	3.05	3.84
Nova 282	BENIGN / NORMAL	2 of 2	2 of 2	3.21	3.50
Nova 318	NORMAL / BENIGN	2 of 2	1 of 3	1.91	1.79
Nova 326	BENIGN / NORMAL	3 of 3	3 of 3	2.00	2.50
Nova 333	BENIGN / NORMAL	2 of 2	1 of 1	3.12	4.06

Patient	Labels (L / R)	Usable scans, L	Usable scans, R	Water:Fat, L	Water:Fat, R
Nova 338	NORMAL / BENIGN	2 of 3	1 of 2	2.13	1.47
Nova 346	BENIGN / NORMAL	3 of 3	3 of 3	2.54	3.38
Nova 350	NORMAL / BENIGN	2 of 3	1 of 2	1.67	2.41
Nova 360	NORMAL / BENIGN	2 of 3	3 of 3	2.11	2.61
Nova 361	BENIGN / NORMAL	3 of 3	1 of 3	1.87	3.55
Nova 363	NORMAL / BENIGN	2 of 3	3 of 3	3.76	1.92
Nova 367	BENIGN / NORMAL	1 of 3	1 of 2	1.00	2.00
Nova 371	NORMAL / BENIGN	2 of 2	1 of 3	1.77	1.11
Nova 378	BENIGN / NORMAL	1 of 2	2 of 2	1.65	2.29
Nova 379	BENIGN / NORMAL	1 of 3	2 of 3	2.58	4.78
Nova 413	NORMAL / BENIGN	2 of 3	3 of 3	1.95	2.56
Nova 414	NORMAL / BENIGN	3 of 3	3 of 3	2.01	1.43
Nova 415	NORMAL / BENIGN	1 of 1	1 of 1	1.55	2.02
Nova 424	NORMAL / BENIGN	2 of 3	1 of 3	2.38	1.67
Nova 430	NORMAL / BENIGN	2 of 3	3 of 3	4.56	4.01
Nova 433	NORMAL / BENIGN	2 of 2	2 of 3	3.33	3.50
Nova 438	BENIGN / NORMAL	3 of 3	3 of 3	4.73	2.09
Nova 445	NORMAL / BENIGN	1 of 3	2 of 3	2.19	1.35
Nova 543	NORMAL / BENIGN	3 of 3	3 of 3	2.30	1.91
Nova 569	NORMAL / BENIGN	2 of 3	3 of 3	2.34	4.43

F.3 CANCER vs BENIGN

5 patients.

```
plot_group(group_novas["CANCER vs BENIGN"], "CANCER vs BENIGN")
```

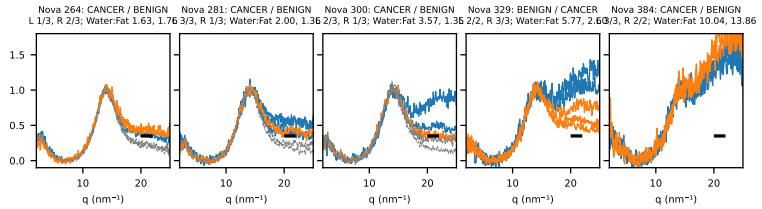


Figure F.3: CANCER vs BENIGN patients with both breasts usable. Same layout as Figure F.1.

Patient	Labels (L / R)	Usable scans, L	Usable scans, R	Water:Fat, L	Water:Fat, R
Nova 264	CANCER / BENIGN	1 of 3	2 of 3	1.63	1.76
Nova 281	CANCER / BENIGN	3 of 3	1 of 3	2.00	1.36
Nova 300	CANCER / BENIGN	2 of 3	1 of 3	3.57	1.35
Nova 329	BENIGN / CANCER	2 of 2	3 of 3	5.77	2.60
Nova 384	CANCER / BENIGN	3 of 3	2 of 2	10.04	13.86

F.4 NORMAL vs NORMAL

2 patients: the healthy baseline.

```
plot_group(group_novas["NORMAL vs NORMAL"], "NORMAL vs NORMAL")
```

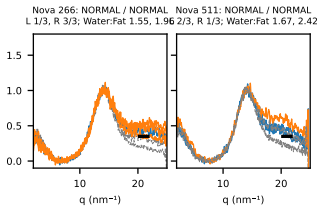


Figure F.4: NORMAL vs NORMAL patients with both breasts usable. Same layout as Figure F.1.

Patient	Labels (L / R)	Usable scans, L	Usable scans, R	Water:Fat, L	Water:Fat, R
Nova 266	NORMAL	1 of 3	3 of 3	1.55	1.96
Nova 511	NORMAL	2 of 3	1 of 3	1.67	2.42

F.5 NORMAL vs NORMAL that missed the threshold

Patients with two NORMAL breasts and **no usable scan in either breast**: every scan sits below the water-peak threshold. Same panel layout; all scans are rejected, so they are shown dashed. 8 patients.

```
plot_missed(nn_missed)
```

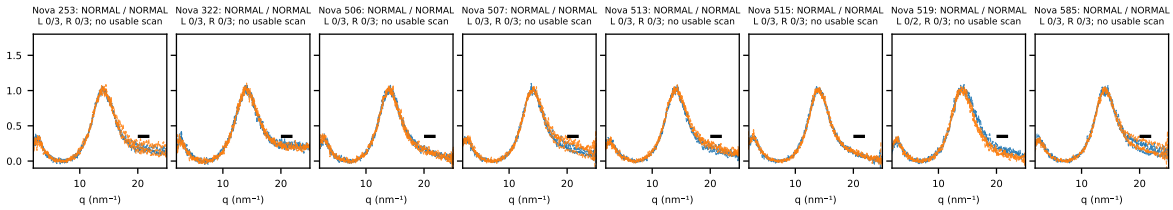


Figure F.5: NORMAL vs NORMAL patients with no usable scan in either breast. Every scan is below the water-peak threshold (black bar); left breast blue dashed, right orange dashed.

Patient	Labels (L / R)	Usable scans, L	Usable scans, R
Nova 253	NORMAL / NORMAL	0 of 3	0 of 3
Nova 322	NORMAL / NORMAL	0 of 3	0 of 3

Patient	Labels (L / R)	Usable scans, L	Usable scans, R
Nova 506	NORMAL / NORMAL	0 of 3	0 of 3
Nova 507	NORMAL / NORMAL	0 of 3	0 of 3
Nova 513	NORMAL / NORMAL	0 of 3	0 of 3
Nova 515	NORMAL / NORMAL	0 of 3	0 of 3
Nova 519	NORMAL / NORMAL	0 of 2	0 of 3
Nova 585	NORMAL / NORMAL	0 of 3	0 of 3

F.6 Other usable pairs

57 patients with both breasts usable whose labels are not one of the pairs above (for example BENIGN/BENIGN, or a side labelled ATYPICAL or PRE_CANCEROUS).

```
plot_group(group_novas["other"], "other")
```

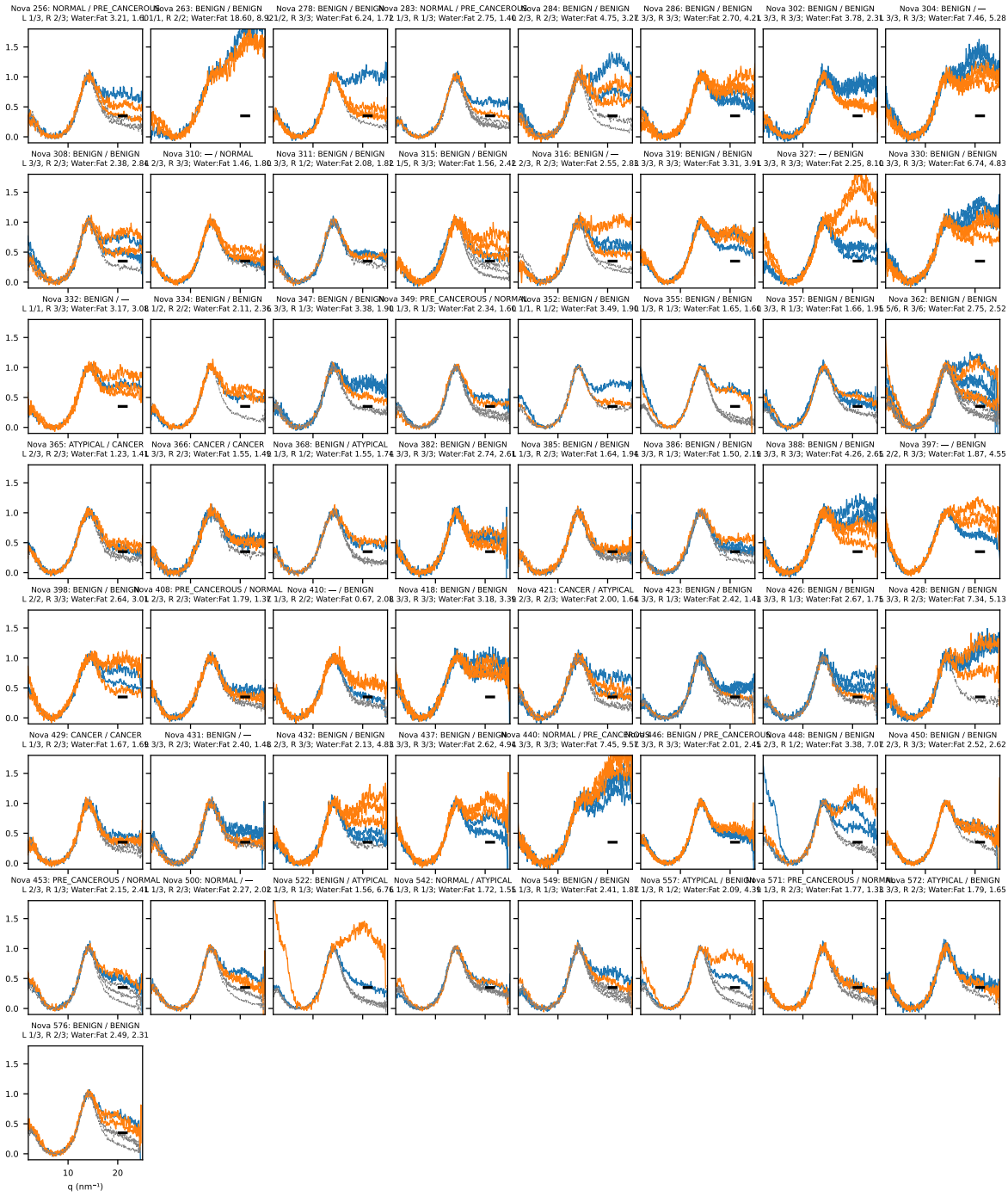


Figure F.6: Other patients with both breasts usable. Same layout as Figure F.1.

Patient	Labels (L / R)	Usable scans, L	Usable scans, R	Water:Fat, L	Water:Fat, R
Nova 256	NORMAL / PRE_CAN-CEROUS	1 of 3	2 of 3	3.21	1.60
Nova 263	BENIGN	1 of 1	2 of 2	18.60	8.92
Nova 278	BENIGN	1 of 2	3 of 3	6.24	1.72
Nova 283	NORMAL / PRE_CAN-CEROUS	1 of 3	1 of 3	2.75	1.40
Nova 284	BENIGN	2 of 3	2 of 3	4.75	3.27
Nova 286	BENIGN	3 of 3	3 of 3	2.70	4.21
Nova 302	BENIGN	3 of 3	3 of 3	3.78	2.31
Nova 304	BENIGN / —	3 of 3	3 of 3	7.46	5.28
Nova 308	BENIGN	3 of 3	2 of 3	2.38	2.84
Nova 310	— / NORMAL	2 of 3	3 of 3	1.46	1.80
Nova 311	BENIGN	3 of 3	1 of 2	2.08	1.82
Nova 315	BENIGN	1 of 5	3 of 3	1.56	2.42
Nova 316	BENIGN / —	2 of 3	2 of 3	2.55	2.83
Nova 319	BENIGN	3 of 3	3 of 3	3.31	3.91
Nova 327	— / BENIGN	3 of 3	3 of 3	2.25	8.10
Nova 330	BENIGN	3 of 3	3 of 3	6.74	4.83
Nova 332	BENIGN / —	1 of 1	3 of 3	3.17	3.08
Nova 334	BENIGN	1 of 2	2 of 2	2.11	2.36
Nova 347	BENIGN	3 of 3	1 of 3	3.38	1.90
Nova 349	PRE_CAN-CEROUS / NORMAL	1 of 3	1 of 3	2.34	1.60
Nova 352	BENIGN	1 of 1	1 of 2	3.49	1.90
Nova 355	BENIGN	1 of 3	1 of 3	1.65	1.60
Nova 357	BENIGN	3 of 3	1 of 3	1.66	1.95
Nova 362	BENIGN	5 of 6	3 of 6	2.75	2.52
Nova 365	ATYPICAL / CANCER	2 of 3	2 of 3	1.23	1.41
Nova 366	CANCER	3 of 3	2 of 3	1.55	1.49
Nova 368	BENIGN / ATYPICAL	1 of 3	1 of 2	1.55	1.74

Patient	Labels (L / R)	Usable scans, L	Usable scans, R	Water:Fat, L	Water:Fat, R
Nova 382	BENIGN	3 of 3	3 of 3	2.74	2.61
Nova 385	BENIGN	1 of 3	2 of 3	1.64	1.94
Nova 386	BENIGN	3 of 3	1 of 3	1.50	2.19
Nova 388	BENIGN	3 of 3	3 of 3	4.26	2.65
Nova 397	— / BENIGN	2 of 2	3 of 3	1.87	4.55
Nova 398	BENIGN	2 of 2	3 of 3	2.64	3.01
Nova 408	PRE_CANCEROUS / NORMAL	2 of 3	2 of 3	1.79	1.37
Nova 410	— / BENIGN	1 of 3	2 of 2	0.67	2.08
Nova 418	BENIGN	3 of 3	3 of 3	3.18	3.39
Nova 421	CANCER / ATYPICAL	2 of 3	2 of 3	2.00	1.64
Nova 423	BENIGN	3 of 3	1 of 3	2.42	1.43
Nova 426	BENIGN	3 of 3	1 of 3	2.67	1.75
Nova 428	BENIGN	3 of 3	2 of 3	7.34	5.13
Nova 429	CANCER	1 of 3	2 of 3	1.67	1.69
Nova 431	BENIGN / —	3 of 3	2 of 3	2.40	1.48
Nova 432	BENIGN	2 of 3	3 of 3	2.13	4.83
Nova 437	BENIGN	3 of 3	3 of 3	2.62	4.94
Nova 440	NORMAL / PRE_CANCEROUS	3 of 3	3 of 3	7.45	9.57
Nova 446	BENIGN / PRE_CANCEROUS	3 of 3	3 of 3	2.01	2.45
Nova 448	BENIGN	2 of 3	1 of 2	3.38	7.07
Nova 450	BENIGN	2 of 3	3 of 3	2.52	2.62
Nova 453	PRE_CANCEROUS / NORMAL	2 of 3	1 of 3	2.15	2.41
Nova 500	NORMAL / —	1 of 3	2 of 3	2.27	2.02
Nova 522	BENIGN / ATYPICAL	1 of 3	1 of 3	1.56	6.76
Nova 542	NORMAL / ATYPICAL	1 of 3	1 of 3	1.72	1.55

Patient	Labels (L / R)	Usable scans, L	Usable scans, R	Water:Fat, L	Water:Fat, R
Nova 549	BENIGN	1 of 3	1 of 3	2.41	1.87
Nova 557	ATYPICAL / BENIGN	1 of 3	1 of 2	2.09	4.39
Nova 571	PRE_CANCEROUS / NORMAL	1 of 3	2 of 3	1.77	1.33
Nova 572	ATYPICAL / BENIGN	3 of 3	2 of 3	1.79	1.65
Nova 576	BENIGN	1 of 3	2 of 3	2.49	2.31

G The newest patients in detail

The newest patients, from Nova 588 to Nova 632, treated exactly like every other patient: each scan is put through the rule of Appendix C (water peak height 0.35 and water centred at q 19), without looking at the labels, most of which are not in yet. Nova 588-595 come from the main archive and Nova 596 onwards from the scans added on 9 November 2026 (measured 2-6 November), prepared in exactly the same way (Figure A.1). Of their 265 scans, 52 (20%) are usable, and **11 of the 45 patients** have both breasts usable. Most labels are not in yet; once they are, the patients with both breasts usable can be used to test the comparison of Appendix C (Chapter 7).

```
COLUMNS = 8
rows_needed = int(np.ceil(len(new_novas) / COLUMNS))
plt.rcParams["font.size"] = 6
fig, axes = plt.subplots(rows_needed, COLUMNS, figsize=(10, 1.45 * rows_needed + 0.3), sharex
for ax, nova in zip(axes.flat, new_novas):
    counts = []
    for side, colour in (("LEFT", "tab:blue"), ("RIGHT", "tab:orange")):
        _, n_good, n_all, _ = breast_summary(nova, side)
        counts.append(f"{side[0]} {n_good}/{n_all}")
        for _, curve, _, _, ok in new_records.get((nova, side), []):
            ax.plot(q, curve, color=colour if ok else "grey", lw=0.8 if ok else 0.6, ls="-" 
    ax.hlines(WATER_MIN, *WATER, color="black", lw=2, zorder=5)
    ax.set_facecolor({"both": BOTH_TINT, "one ": ONE_TINT}.get(outcomes[nova][:4], NONE_TINT)
    ax.set_title(f"Nova {nova}: {', '.join(counts)}", fontsize=6)
    ax.set(xlim=(2, 25), ylim=(-0.1, 1.8))
for ax in axes.flat[len(new_novas):]:
    ax.axis("off")
for ax in axes[-1]:
    ax.set_xlabel("q (nm ⁻¹)")
fig.tight_layout(pad=0.4)
plt.show()
plt.rcParams["font.size"] = 10
```

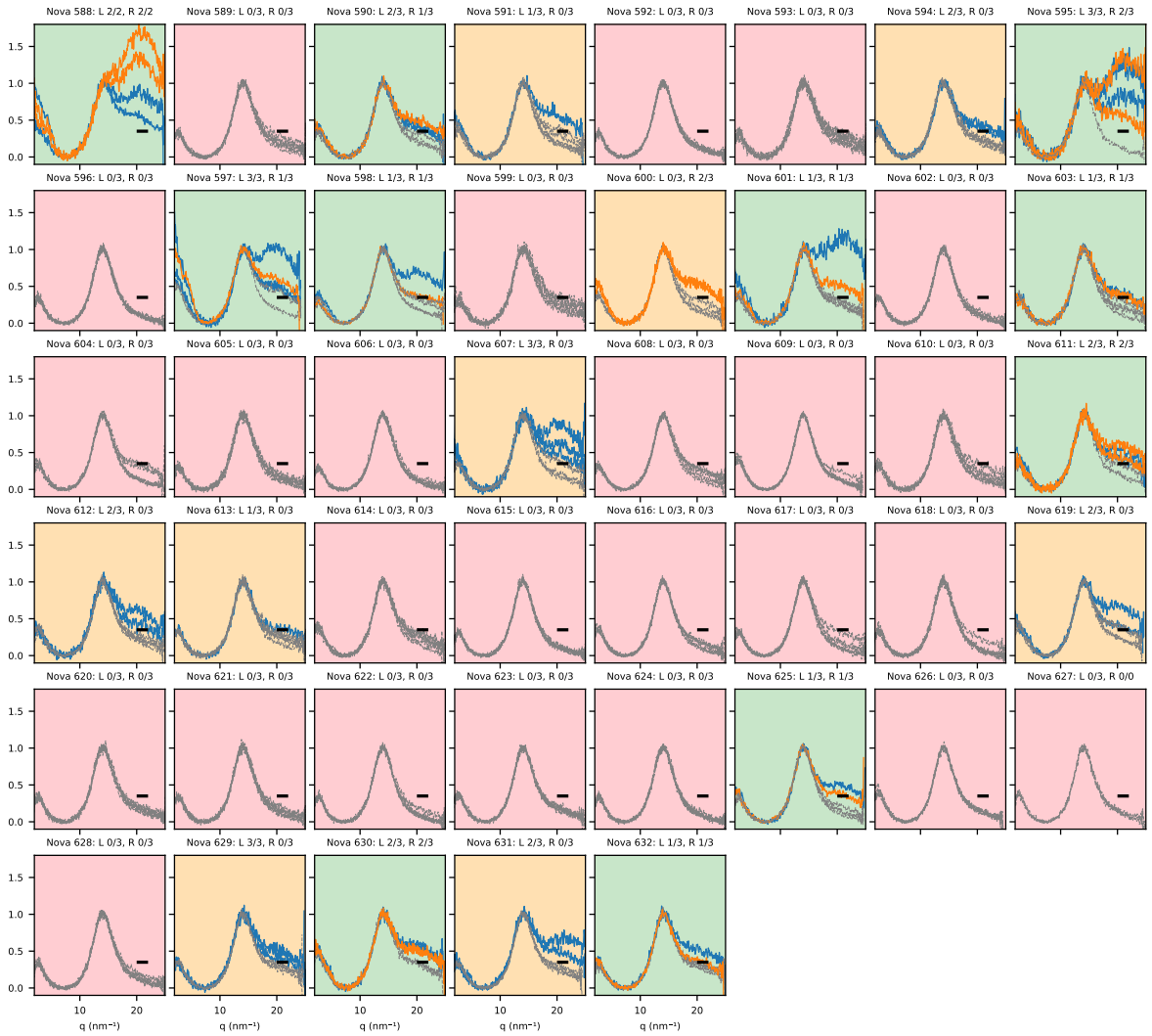


Figure G.1: Every prepared scan of the newest patients, one panel per patient, tinted by the outcome as in the table (green both breasts usable, orange one, red neither). Usable scans are coloured (left breast blue, right orange), scans the rule rejects are grey and dashed; the black bar is the threshold on the water peak (q 20-22).

Patient	Labels (L / R)	Usable scans, L	Usable scans, R	Water:Fat, L	Water:Fat, R
Nova 588	NORMAL / CANCER	2 of 2	2 of 2	3.13	9.08
Nova 589	not yet labelled	0 of 3	0 of 3	–	–
Nova 590	not yet labelled	2 of 3	1 of 3	1.63	2.07
Nova 591	not yet labelled	1 of 3	0 of 3	1.86	–
Nova 592	not yet labelled	0 of 3	0 of 3	–	–

Patient	Labels (L / R)	Usable scans, L	Usable scans, R	Water:Fat, L	Water:Fat, R
Nova 593	not yet labelled	0 of 3	0 of 3	–	–
Nova 594	not yet labelled	2 of 3	0 of 3	1.80	–
Nova 595	ATYPICAL	3 of 3	2 of 3	4.75	4.90
Nova 596	NORMAL / not yet labelled	0 of 3	0 of 3	–	–
Nova 597	not yet labelled	3 of 3	1 of 3	2.69	2.52
Nova 598	not yet labelled	1 of 3	1 of 3	3.60	1.60
Nova 599	not yet labelled	0 of 3	0 of 3	–	–
Nova 600	not yet labelled	0 of 3	2 of 3	–	2.09
Nova 601	not yet labelled	1 of 3	1 of 3	5.13	2.21
Nova 602	not yet labelled	0 of 3	0 of 3	–	–
Nova 603	not yet labelled	1 of 3	1 of 3	1.62	1.21
Nova 604	ATYPICAL	0 of 3	0 of 3	–	–
Nova 605	not yet labelled	0 of 3	0 of 3	–	–
Nova 606	not yet labelled	0 of 3	0 of 3	–	–
Nova 607	not yet labelled	3 of 3	0 of 3	2.73	–
Nova 608	BENIGN / ATYPICAL	0 of 3	0 of 3	–	–
Nova 609	NORMAL / CANCER	0 of 3	0 of 3	–	–
Nova 610	not yet labelled	0 of 3	0 of 3	–	–
Nova 611	not yet labelled	2 of 3	2 of 3	1.77	1.99
Nova 612	BENIGN	2 of 3	0 of 3	1.75	–
Nova 613	not yet labelled	1 of 3	0 of 3	1.34	–
Nova 614	not yet labelled	0 of 3	0 of 3	–	–
Nova 615	not yet labelled	0 of 3	0 of 3	–	–
Nova 616	not yet labelled	0 of 3	0 of 3	–	–
Nova 617	not yet labelled	0 of 3	0 of 3	–	–
Nova 618	BENIGN / NORMAL	0 of 3	0 of 3	–	–
Nova 619	not yet labelled / BENIGN	2 of 3	0 of 3	2.25	–
Nova 620	not yet labelled	0 of 3	0 of 3	–	–
Nova 621	not yet labelled	0 of 3	0 of 3	–	–
Nova 622	not yet labelled	0 of 3	0 of 3	–	–
Nova 623	not yet labelled	0 of 3	0 of 3	–	–
Nova 624	not yet labelled	0 of 3	0 of 3	–	–
Nova 625	not yet labelled	1 of 3	1 of 3	2.08	1.57
Nova 626	not yet labelled	0 of 3	0 of 3	–	–
Nova 627	not yet labelled / not scanned	0 of 3	0 of 0	–	–
Nova 628	not yet labelled	0 of 3	0 of 3	–	–
Nova 629	not yet labelled	3 of 3	0 of 3	1.84	–
Nova 630	not yet labelled	2 of 3	2 of 3	2.39	2.00
Nova 631	not yet labelled	2 of 3	0 of 3	2.40	–
Nova 632	NORMAL / ATYPICAL	1 of 3	1 of 3	1.97	1.38

Water:Fat is worked out, as in Appendix C, from the mean of each breast's usable scans only. The patients with neither breast usable have fat-dominated scans throughout, as most patients the rule leaves out do (Figure D.6). Once their labels come in, the patients compared can be added to the comparison of Appendix C (Chapter 7).

H Tumour size by grade

The size of each tumour in our CANCER breasts, by grade, measured two ways. From the pathology reports, the **linear extent** of invasive tumour measured in the biopsy cores (the largest of a breast's specimens, lymph node specimens left out); from the radiology reports, the **largest mass measured on imaging** (mammogram, ultrasound or MRI). The grade is the **overall (Nottingham) grade** from the pathology reports, taken from the score out of 9 where the report gives one (3-5 grade 1, 6-7 grade 2, 8-9 grade 3) and from the stated grade otherwise. The imaging sizes are taken from the report text and can include more than one mass in a breast, so they are approximate.

```
GRADES = (1, 2, 3)
rng = np.random.default_rng(0)  # the same horizontal jitter on every render

def plot_by_grade(ax, values, ylabel, title):
    by_grade = {g: [size for grade, size, _ in values if grade == g] for g in GRADES}
    ax.boxplot([by_grade[g] for g in GRADES], positions=range(len(GRADES)), widths=0.5, show
               boxprops=dict(facecolor="#d4ebe8", edgecolor="#2a9d8f"), whiskerprops=dict(co
               medianprops=dict(color="#1d3557", lw=2.2))
    for k, g in enumerate(GRADES):
        for is_usable in (False, True):
            here = [size for grade, size, ok in values if grade == g and ok == is_usable]
            ax.scatter(k + rng.uniform(-0.18, 0.18, len(here)), here, s=14 if is_usable else
                       color=TOP_COLOUR if is_usable else "grey", alpha=0.9 if is_usable else
                       edgecolor="black" if is_usable else "none", lw=0.4,
                       label=("usable scan" if is_usable else "no usable scan") if k == 0 el
    labels = []
    for g in GRADES:
        here = [(size, ok) for grade, size, ok in values if grade == g]
        labels.append(f"Grade {g}\nn={len(here)}\nmean {np.mean([s for s, _ in here]):.1f} m
                      if here else f"Grade {g}\n(n=0)")
    ax.set_xticks(range(len(GRADES)), labels)
    ax.set(ylabel=ylabel, title=title, ylim=(0, None))
    ax.grid(alpha=0.3, axis="y")
    ax.legend(frameon=False, loc="upper left")

plt.rcParams["font.size"] = 7
```

```
fig, (biopsy_ax, imaging_ax) = plt.subplots(1, 2, figsize=(8, 3.8))
plot_by_grade(biopsy_ax, tumours, "Linear extent (mm)", "Biopsy cores")
plot_by_grade(imaging_ax, imaged, "Largest mass on imaging (mm)", "Imaging")
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

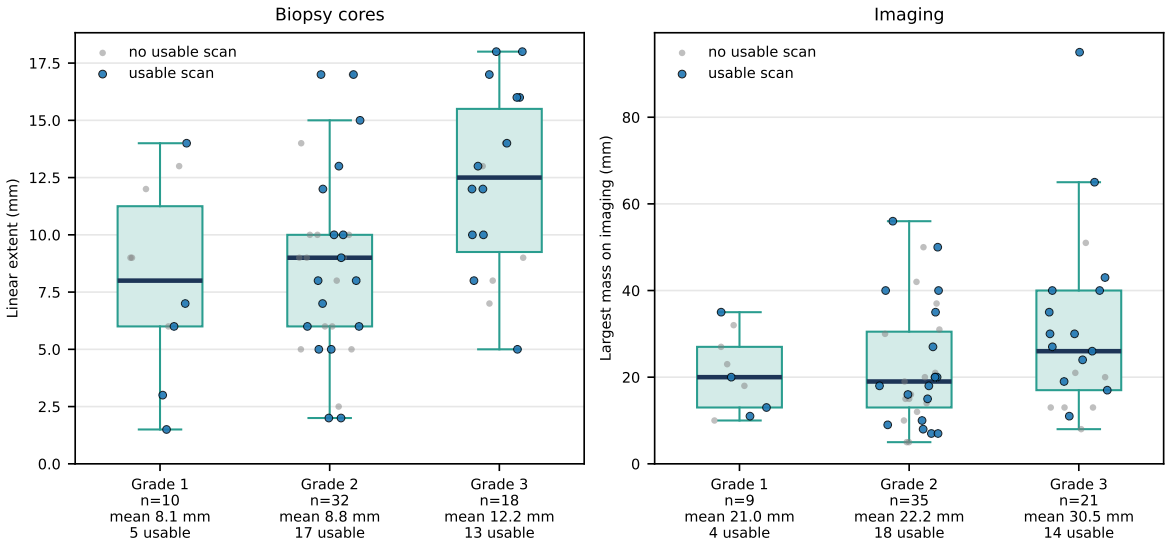


Figure H.1: Tumour size of each CANCER breast by overall grade; one dot per breast. **Left**: the linear extent of invasive tumour in the biopsy cores (pathology reports). **Right**: the largest mass measured on imaging (mammogram, ultrasound or MRI reports). Boxes span the middle half of the breasts, the thick line is the median, and the whiskers reach the furthest breast within 1.5 box-lengths. Breasts with at least one usable scan are blue, the others grey.

I Excluding patients with a large left/right intensity imbalance

`8 patients with image-total imbalance  2× excluded; 251 left`

A few patients have one breast whose raw images are far brighter than the other (total detector counts), even at similar thickness. That imbalance is rare in Nova 250 onwards (typical left/right ratio near 1) and produces unreliable Water:Fat comparisons — Nova 396 is the example in Figure I.1, where the CANCER side's mean image total is about 2.4× the NORMAL side's and the NORMAL curve looks almost fat-free. From here on, a patient is dropped if the brighter breast's mean image total is at least **2×** the dimmer breast's (`breast_image_totals.csv`, summed over each Sample Main image before 1D reduction). That removes **8** patients, including Nova 396; 7 of them are among the 258 patients of this analysis, leaving 251.

```
EXCLUDE_EXAMPLE = 396
exclude_totals = {row.side: float(row.image_total_mean) for row in totals[totals.nova == EXCL
exclude_scans = {}  # side: [(label, curve, water height)]
for side, label, scan_q, curve, _ in ne.read_patients(ne.STORE_PATH, {EXCLUDE_EXAMPLE}, ne.RE
    on_q = np.interp(q, scan_q, curve, left=np.nan, right=np.nan)
    with np.errstate(all="ignore"):
        height = np.nanmean(on_q[(q >= WATER[0]) & (q <= WATER[1])])
    exclude_scans.setdefault(side, []).append((label, on_q, height))

plt.rcParams["font.size"] = 7
fig, axes = plt.subplots(1, 2, figsize=(8, 4.3), sharey=True)
x = np.linspace(2, 25, 600)
sides = sorted(exclude_scans, key=lambda side: exclude_scans[side][0][0] != "NORMAL")
for ax, side in zip(axes, sides):
    ax.set_box_aspect(1)
    items = exclude_scans[side]
    label = items[0][0]
    curves = [curve for _, curve, _ in items]
    with warnings.catch_warnings():
        warnings.simplefilter("ignore", RuntimeWarning)
        mean = np.nanmean(curves, axis=0)
    p = fit_two_gaussians(mean)
```

```
    ratio = water_fat(mean)
    for k, (_, curve, height) in enumerate(items):
        ax.plot(q, curve, color=ne.STATUS_COLOURS[label], lw=0.9, label=f"scan {k + 1}: wate
    for (height, centre, sigma), colour in ((p[:3], FAT_COLOUR), (p[3:], WATER_COLOUR)):
        within = (x >= centre - 3 * sigma) & (x <= centre + 3 * sigma)
        ax.fill_between(x[within], gaussian(x[within], height, centre, sigma), color=colour,
    ax.plot(x, two_gaussians(x, *p), color="red", ls="--", lw=1.4, zorder=5, label="fat + wa
    img_total = exclude_totals.get(side, np.nan)
    ax.set(title=f"Nova {EXCLUDE_EXAMPLE} {side}, {label}: image total {img_total:.2g}\n"
                 f"Water:Fat = {ratio:.2f}  (patient excluded: imbalance {intensity_ratio[E
           xlabel="q (nm ⁻¹)", xlim=(2, 25), ylim=(0, 2))
    ax.set_xticks([5, 10, 15, 20, 25])
    ax.grid(alpha=0.3)
    ax.legend(frameon=False, fontsize=6, loc="upper left")
axes[0].set_ylabel("I(q), stretched")
fig.tight_layout()
plt.show()
plt.rcParams["font.size"] = 10
```

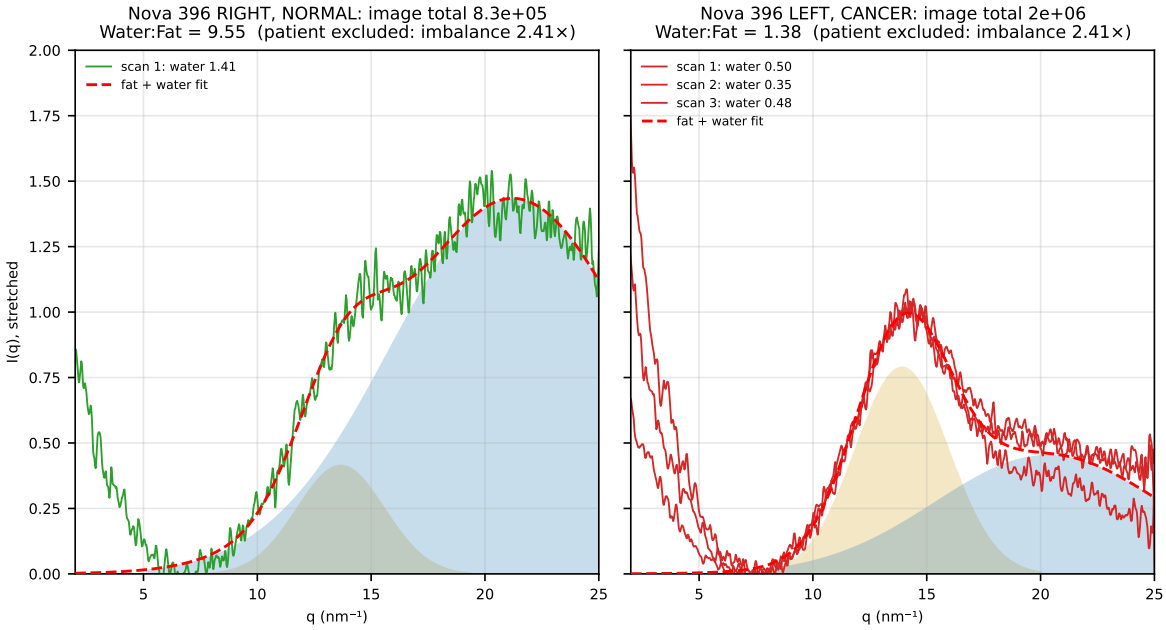


Figure I.1: An excluded patient: Nova 396. Mean image total on the CANCER (LEFT) side is 2.4× the NORMAL (RIGHT) side. **Left**: NORMAL scans after preparation — water-dominated, weak fat peak. **Right**: CANCER scans with a clear fat peak. Patients whose two breasts differ by 2× in mean image total are dropped before the Water:Fat comparisons.